Periods of GeoVac: a cyclotomic mixed-Tate classification of the master Mellin engine

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Abstract

The GeoVac framework is a discrete spectral-graph approach to quantum mechanics built around the Fock conformal projection $\mathbb{R}^3 \hookrightarrow S^3$. Its computational observables empirically produce a constrained class of transcendental numbers: π and integer powers thereof, classical odd zeta values $\zeta(2k+1)$, Catalan $G = \beta(2)$ and the Dirichlet beta values $\beta(s) = L(s, \chi_{-4})$, and a depth-$k$ tower of constants at higher loop order, identified in closed form at $k = 2$ (Paper 28 of the GeoVac corpus). We prove that every such transcendental sits in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$, with a complete period-theoretic stratification indexed by the operator-order slot $k \in \{0, 1, 2\}$ of the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ on the discrete Camporesi–Higuchi S^3 spectral triple:

- $k = 0$ (Hopf-base measure $\mathrm{Vol}(S^2)/4 = \pi$): $M_1 \subset \mathbb{Q}[\pi, \pi^{-1}]$, the localised pure-Tate algebra at π, depth 0.
- $k = 2$ (Seeley–DeWitt coefficients): $M_2 \subset \bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, the pure-Tate even-weight sub-ring, strictly smaller than the generic Fathizadeh–Marcolli [25] mixed-Tate classification of Robertson–Walker spectral actions: on the static S^3 sub-case, the Dirac SD expansion is two-term exact and the scalar Laplacian SD coefficients form a closed geometric series.
- $k = 1$ (vertex parity / Hurwitz at quarter-integer shifts / Dirichlet L): $M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4 via Deligne [23] and the explicit basis of Glanois [24], at depth equal to loop order. The vertex-parity-as-χ_{-4} correspondence is the Deligne–Glanois Galois descent from level 4 to level 2 realised on the framework's natural QED observables.

The three sub-mechanisms are not set-theoretically disjoint (joint engagement is the typical case in physical observables); however, the master Mellin index k uniquely identifies the period-ring contribution of each spectral object. We catalogue witnesses from across the corpus (Papers 2, 25, 28, 38, 40, 43, 50, 51) and indicate open questions: $S_{\min}$ is now fully resolved — both remaining Hoffman t-values are identified and $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ in explicit closed form (Paper 28; the prior "PSLQ-irreducible" label was a coverage artifact, corrected 2026-06-11); inhomogeneous extension to non-static Robertson–Walker substrates; cross-manifold extension to the Bargmann–Segal Hardy lattice on S^5; and inner-factor Yukawa coupling content.

Contents

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1 Introduction

1.1 Background: GeoVac and the transcendental-to-periods question

The GeoVac framework [7, 8] treats the S^3 conformal projection of the hydrogen Coulomb problem (Fock 1935 [19]) as a discrete spectral-graph object: nodes labeled by quantum numbers $(n, l, m, \ldots)$, edges by spectral selection rules, and a (positive-semidefinite) graph Laplacian whose spectrum converges, under the Fock projection, to the continuum Coulomb spectrum, with $\kappa = -1/16$ the matching scaling constant (an Observation; Paper 7). The framework has been extended into a multi-paper corpus covering quantum chemistry, qubit encodings, gauge structure, Lorentzian extension, and gravity. A consistent feature throughout: the *algebraic skeleton* of the framework (the bare graph and its combinatorial selection rules) produces matrix elements in $\mathbb{Q}$, while *transcendentals* enter at projection steps that promote the discrete substrate to continuum observables. Paper 18 of the corpus [8] catalogues these as "exchange constants" organised by six tiers (intrinsic, calibration, embedding, algebraic-implicit, composition, inner-factor input data).

The empirical content of the catalogue is short: across the entire GeoVac corpus, only the following transcendentals have ever appeared in spectral data:

$$\pi^n \, (n \in \mathbb{Z}), \quad \zeta(2k+1) \, (k \geq 1), \quad G = \beta(2), \quad \beta(4), \; \beta(s), \; \ldots, \quad \text{plus a depth-}k\text{ tower} \quad S_{\min}, S^{(3)}, \ldots \tag{1}$$

(Note: $S^{(4)}$ appeared in an earlier version of this equation but was never defined, computed, or tested anywhere in the corpus; it has been removed.)

The first three classes have classical period-theoretic identifications (Kontsevich–Zagier 2001 [21]; Brown 2012 [22]; Deligne 2010 [23]; Glanois 2015 [24]). The depth-k tower constants are identified elements of $\mathcal{MT}(\mathbb{Z}[1/2])$: $S_{\min}$ decomposes exactly into Hoffman double t-values [30] (classical level-2 alternating MZVs; Paper 28 §$S_{\min}$ [11], round-2 dossier). The prior "PSLQ-irreducible" label was a basis-coverage artifact (the tested bases contained ≤ 2 of the 8 weight-5 level-2 monomials required).

The natural question, opened by the empirical content of Eq. (1): do all of GeoVac's transcendentals sit in a single, named period-theoretic class? If so, what is the structural mechanism that selects this class?

1.2 Result: master Mellin engine partition is a complete period-ring stratification

We show that the answer is affirmative. Every transcendental in Eq. (1) lies in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$, with a complete stratification indexed by the operator-order slot $k \in \{0, 1, 2\}$ of the *master Mellin engine*

$$\mathcal{M}\Big[\operatorname{Tr}(D^k \, e^{-tD^2})\Big](s) \tag{2}$$

on the discrete Camporesi–Higuchi S^3 spectral triple [20]. The structure is:

k	Sub-mechanism	Period-ring home	Source
0	M1 (Hopf-base measure)	$\mathbb{Q}[\pi, \pi^{-1}]$	this paper §3
2	M2 (Seeley–DeWitt)	$\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ pure Tate	[25] + this paper §4
1	M3 (vertex parity / Hurwitz / Dirichlet L)	$\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4	[23, 24] + this paper §5

The M3 row contains the substantive new identification: the vertex-parity selection rule that produces Catalan G and $\beta(s)$ values in the framework's two-loop QED sector (Paper 28 Theorem 3, [11]) is precisely the Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural Dirichlet series $D(s) = D_{\text{even}}(s) + D_{\text{odd}}(s)$, with Hurwitz at quarter-integer shifts $\zeta(s, 1/4), \zeta(s, 3/4)$ surviving in $D_{\text{even}} - D_{\text{odd}}$ but cancelling in the sum.

1.3 Structure

Section 2 sets up the master Mellin engine on the Camporesi–Higuchi spectrum and states the case-exhaustion theorem of Paper 32 § VIII [13]. Sections 3, 4, and 5 prove the three sub-mechanism classifications, each followed by a witness catalogue from the corpus. Section 6 treats joint engagement and physical observables that engage multiple sub-mechanisms simultaneously (the canonical example being the K-formula $K = \pi(B + F - \Delta)$ of Paper 2 [6]). Section 7 catalogues open questions: specific motivic identification of the depth-k tower, inhomogeneous extension, cross-manifold extension to the S^5 Bargmann–Segal Hardy lattice [9], and inner-factor Yukawa content [13]. Section 8 concludes.

2 The master Mellin engine on Camporesi–Higuchi S^3

2.1 The Camporesi–Higuchi spectrum

Let D_{CH} denote the canonical Dirac operator on the round S^3 of unit radius, with the Camporesi–Higuchi spectral realisation [20]:

$$|\lambda_n^{\mathrm{CH}}| = n + \tfrac{3}{2}, \qquad g_n = 2(n+1)(n+2), \qquad n \in \mathbb{N}_{\geq 0}, \tag{3}$$

where the half-integer spectrum arises from the spinor parallel transport on the parallelisable group manifold $S^3 = \mathrm{SU}(2)$. The squared Dirac spectrum is the integer-coefficient quadratic $\lambda_n^2 = (n + 3/2)^2$ with the same degeneracy. The associated spectral zeta function is

$$D(s) := \sum_{n \geq 0} \frac{g_n}{|\lambda_n^{\mathrm{CH}}|^s} = 2\,\zeta(s-2, 3/2) - \tfrac{1}{2}\,\zeta(s, 3/2), \tag{4}$$

where $\zeta(s, a)$ is the Hurwitz zeta function, and the squared spectral zeta is

$$\zeta_{D^2}(s) := \sum_{n \geq 0} \frac{g_n}{(\lambda_n^{\mathrm{CH}})^{2s}} = D(2s). \tag{5}$$

2.2 The master Mellin engine

The Mellin transform of the trace $\mathrm{Tr}(D^k\, e^{-tD^2})$ extracts the spectral zeta value $D(2s - k)$ (modulo Γ-factor normalisations):

$$\mathcal{M}\Big[\mathrm{Tr}(D^k\, e^{-tD^2})\Big]\,(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{n \geq 0} g_n (\lambda_n)^k\, e^{-t\lambda_n^2}\, dt = \frac{\Gamma(s + k/2)}{\Gamma(s)} \cdot D(2s + k). \tag{6}$$

At integer s, the three cases $k \in \{0, 1, 2\}$ produce structurally distinct outputs:

- $k = 0$: trivial heat kernel collapses to volume integral over the substrate; the leading $t \to 0^+$ coefficient is $\mathrm{Vol}(S^3)/(4\pi)^{3/2}$ via the Weyl asymptotic. The Hopf-bundle decomposition $S^3 \to S^2$ (Paper 25, [10]) gives the base measure factor $\mathrm{Vol}(S^2)/4 = \pi$ in the explicit identifications below; we designate this contribution as mechanism M1.

- $k = 2$: standard Connes–Chamseddine spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ produces Seeley–DeWitt coefficients $\{a_k(D^2)\}_{k \geq 0}$ via the asymptotic expansion of the heat trace as $t \to 0^+$. We designate this contribution as M2.

- $k = 1$: the η-invariant series $\eta(s) := \sum_n \mathrm{sgn}(\lambda_n) g_n |\lambda_n|^{-s}$ arises by Mellin transform of $\mathrm{Tr}(De^{-tD^2})$, with the vertex-parity decomposition $D = D_{\mathrm{even}} + D_{\mathrm{odd}}$ on the parity of n_{Fock} exposing Hurwitz at quarter-integer shifts. We designate this contribution as M3.

2.3 The case-exhaustion theorem

Paper 32 § VIII Theorem 2.1 ([13]) establishes:

Theorem 2.1 (Case-exhaustion of π-sources on the discrete S^3 spectral triple)**.** *Every appearance of π in any closed-form GeoVac spectral observable arises from one (or more) of the three named sub-mechanisms M_1, M_2, M_3:*

- *M_1: Hopf-base measure factor $\mathrm{Vol}(S^2)/4 = \pi$ on the Hopf bundle $S^3 \to S^2$;*
- *M_2: Seeley–DeWitt heat-kernel coefficient $a_k(D^2) \in \pi \cdot \mathbb{Q}$ in the asymptotic expansion of $\mathrm{Tr}\, e^{-tD^2}$;*
- *M_3: vertex-restricted Dirichlet-L content via Hurwitz at quarter-integer shifts in the parity decomposition of $D(s)$.*

A falsification sweep against three candidate fourth mechanisms (continuum gravity, anomaly classes, principal bundles) returned the sharper finding that M_1, M_2, M_3 are three sub-cases of the single master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ parameterised by operator order $k \in \{0, 1, 2\}$.

The theorem partitions GeoVac's π-content by Mellin slot; it does not by itself classify the *period-theoretic ring* in which each sub-mechanism's output lives. That classification — which is the content of the present paper — requires separate motivic input for each slot, drawn from the published period theory of Kontsevich–Zagier [21], Fathizadeh–Marcolli [25], Deligne [23], and Glanois [24].

3 M1: the Hopf-base measure sub-mechanism

3.1 Period-ring statement

Theorem 3.1 (M1 period-ring)**.** *The M1 sub-mechanism on the discrete Camporesi–Higuchi S^3 spectral triple produces period values in the localised pure-Tate sub-ring*

$$M_1 \;\subset\; \mathbb{Q}[\pi, \pi^{-1}] \tag{7}$$

of mixed-Tate periods over $\mathbb{Q}$ (Kontsevich–Zagier 2001 [21]), at depth 0 and Tate weight $\in \mathbb{Z}$.

Proof. By construction, the M1 mechanism is the Hopf-base measure factor $\mathrm{Vol}(S^2)/4 = \pi$ on the Hopf bundle $S^3 \to S^2$ (Paper 25 § II.1, [10]). As a period in the Kontsevich–Zagier framework, π has the canonical representation $\pi = \frac{1}{2i}\oint_{|z|=1} dz/z$, a Tate weight-1 period of the mixed Tate motive $\mathbb{Q}(1)$. Every M1 closed form across the corpus is a $\mathbb{Q}$-linear combination of integer powers of this period (witnesses tabulated below); no M1 closed form has produced transcendentals outside $\mathbb{Q}[\pi, \pi^{-1}]$. The depth-0 property follows from the absence of iterated integration in the Hopf-measure step: M1 acts as a single multiplicative factor, not as a Mellin convolution. The localisation at π^{-1} accommodates the inverse-period witnesses $4/\pi$ and $1/\pi^2$, which are not periods in the strict Kontsevich–Zagier sense but sit naturally in the field generated by π. □

3.2 Witnesses

Witness	Source	M1 factor	Tate weight
$\mathrm{Vol}(S^2)/4 = \pi$	Paper 25 § II.1 [10] (Hopf base)	π	$+1$
$4/\pi$ (asymptotic state-space GH rate)	Paper 38 § L2 [14]	π^{-1}	-1
$4/\pi$ (universal across compact Lie groups)	Paper 40 Thm 1 [15]	π^{-1}	-1
$1/\pi^2$ (Pythagorean orthogonality prefactor)	Paper 43 § 10.2 [16]	π^{-2}	-2
$1/\pi^2$ (F-theorem cross-product)	Paper 50 Thm 3.4 [17]	π^{-2}	-2

Every entry is $q \cdot \pi^n$ for $q \in \mathbb{Q}$ and $n \in \mathbb{Z}$ with $|n| \leq 2$ on the empirical witnesses logged so far. The K-formula $K = \pi(B + F - \Delta)$ of Paper 2 [6] contains the canonical M1 factor (the leading π); the analysis of joint engagement in §6 below decomposes the full M1+M2+M3 content of the K-formula.

3.3 Convention-freeness

Unlike M2 (which depends on the volume-normalisation convention for the spectral action — see §4 below), the M1 ring is convention-free. The reason: the Hopf-base measure is itself a volume integral, not a heat-kernel asymptotic coefficient at $t \to 0^+$. There is no choice of dimensional rescaling to absorb or expose; the factor π appears directly. This convention-freeness is the structural counterpart of M1's depth-0 status: depth ≥ 1 mechanisms admit non-trivial rescaling choices via the heat-kernel asymptotic, while depth-0 mechanisms do not.

4 M2: the Seeley–DeWitt sub-mechanism

4.1 Period-ring statement

Theorem 4.1 (M2 period-ring)**.** *The M2 sub-mechanism on the discrete Camporesi–Higuchi S^3 spectral triple produces period values in the strictly smaller pure-Tate even-weight sub-ring*

$$M_2 \;\subset\; \bigoplus_{k\geq 0} \pi^{2k} \cdot \mathbb{Q} \;\subset\; \mathcal{MT}(\mathbb{Q}) \tag{8}$$

of mixed-Tate periods over $\mathbb{Q}$, at depth 0. The classification inherits the Fathizadeh–Marcolli [25] mixed-Tate result for Connes–Chamseddine spectral actions on Robertson–Walker spacetimes $\mathbb{R} \times S^3$, restricted to the static sub-case $a(t) \equiv 1$, and refines it in two structural directions: (i) the Dirac SD expansion is two-term exact ($a_k = 0$ for $k \geq 2$), while the F–M generic R–W expansion is infinite; (ii) the scalar Laplacian SD coefficients form the closed geometric series $a_k^{\Delta} = 2\pi^2/k!$, while the F–M generic R–W scalar expansion has no such closed form.

Proof sketch. The inheritance is via specialisation of the Fathizadeh–Marcolli [25] Theorem 6.2: on the general Robertson–Walker spacetime $\mathbb{R} \times S^3$ with metric $dt^2 + a(t)^2 d\Omega_3^2$, the a_{2n} Seeley–DeWitt coefficient (prior to time integration) is a period of an algebraic differential form $\omega_{\varepsilon_1,\ldots,\varepsilon_{2n}}(u_0, u_1, \ldots, u_{2n+2})$ defined on the affine complement

$$\mathcal{V}_n^{\text{F-M}}(\lambda) \;:=\; \mathbb{A}^{2n+3} \;\setminus\; (C_{Z_{\lambda,2n}} \,\cup\, H_0 \,\cup\, H_1), \tag{9}$$

where $H_0 = \{u_0 = 0\}$, $H_1 = \{u_0 = 1\}$, $C_{Z_{\lambda,2n}}$ is the affine cone of the projective quadric defined by

$$Q_{\lambda,2n}(u_1, \ldots, u_{2n+2}) \;=\; u_1^2 + \lambda^{-2}(u_2^2 + u_3^2 + u_4^2) + u_5^2 + u_6^2 + \ldots + u_{2n+2}^2, \tag{10}$$

$\lambda \in \mathbb{G}_m$ is the affine parameterisation of the scaling factor $a(t)$, and $\varepsilon_i = a^{(i)}(t)$ for $i = 1, \ldots, 2n$ are the auxiliary affine parameters corresponding to the time derivatives of the scaling factor. The algebraic differential form $\omega_{\varepsilon_1,\ldots,\varepsilon_{2n}}$ is polynomial in $\varepsilon_1, \ldots, \varepsilon_{2n}$ with monomials carrying explicit exponents $k_1, \ldots, k_{2n}$ in the expansion of Fathizadeh–Marcolli Theorem 6.1.

The GeoVac static sub-case $a(t) \equiv 1$ corresponds to specialisation along the closed immersion

$$\iota : \; \{1\} \times \{0\}^{2n} \;\hookrightarrow\; \mathbb{G}_m \times \mathbb{A}^{2n}_{(\varepsilon_1,\ldots,\varepsilon_{2n})}, \qquad \lambda = 1, \;\; \varepsilon_1 = \ldots = \varepsilon_{2n} = 0. \tag{11}$$

On this slice, every monomial of the Fathizadeh–Marcolli integrand that carries an ε-factor vanishes identically; the surviving "ε-independent core" depends only on $u_0, u_1, \ldots, u_{2n+2}$ with all u_i exponents even, and the quadric specialises to the standard unit-isotropic form

$$Q_{1,2n} \;=\; u_1^2 + u_2^2 + u_3^2 + u_4^2 + u_5^2 + \ldots + u_{2n+2}^2. \tag{12}$$

The corresponding GeoVac affine complement is

$$\mathcal{V}_n^{\mathrm{GeoVac}} := \mathbb{A}^{2n+3} \setminus (C_{Z_{1,2n}} \cup H_0 \cup H_1). \tag{13}$$

Over the quadratic extension $\mathbb{Q}(\sqrt{-1})$, the quadric $Q_{1,2n}$ is hyperbolic (isotropic), and the motive of $\mathcal{V}_n^{\mathrm{GeoVac}}$ is mixed Tate by direct application of Fathizadeh–Marcolli Theorem 7.3 (at $n = 1$), Theorem 7.5 (at $n = 2$), and the inductive treatment of § 7.6, 7.7 (general n). Over $\mathbb{Q}$, the period output lies in the pure-Tate even-weight sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$.

The pure-Tate refinement on the static sub-case has two parts at the spectrum level:

- *Dirac sector.* The Camporesi–Higuchi spectrum $|\lambda_n| = n + 3/2$ admits the Jacobi ϑ_2 modular transformation $\mathrm{Tr} e^{-tD^2} = (\sqrt{\pi}/2)t^{-3/2} - (\sqrt{\pi}/4)t^{-1/2} + O(e^{-\pi^2/t})$, with the exponentially small remainder vanishing in the asymptotic series. All SD coefficients $a_k^{D^2}$ for $k \geq 2$ are exactly zero; the volume-normalised coefficients are $a_0 = 4\pi^2$, $a_1 = -2\pi^2$.

- *Scalar sector.* The integer spectrum $n(n+2)$ admits the Jacobi ϑ_3 modular transformation giving the closed form $\mathrm{Tr} e^{-t\Delta} = (\sqrt{\pi}/4) \cdot e^t/t^{3/2} + O(e^{-\pi^2/t})$, with $a_k^{\Delta} = 2\pi^2/k!$.

Both sectors are explicit in Paper 51 [18]. The empirical pure-Tate restriction (no $\zeta(3)$, no MZVs) is consistent with the motivic collapse: the F–M generic R–W classification admits MZV content through the ε_i dependence (time derivatives of the scaling factor); on the static sub-case those derivatives vanish, eliminating the MZV content at the motivic level.

A precise refinement of the "trivial complement" reading: the (λ, ε)-fiber of the F–M classification collapses to the single point $\{1\} \times \{0\}^{2n}$ on the GeoVac sub-case; the u-coordinate complement $\mathcal{V}_n^{\mathrm{GeoVac}}$ remains a genuine $(2n + 3)$-dimensional mixed-Tate variety. □

Remark 4.2 (Specialisation as motive equivalence)**.** The relationship between the F–M generic Robertson–Walker classification and the GeoVac static S^3 specialisation is a closed immersion of relative motives: the F–M classification supplies the universal "period-with-parameter-data" object $\mathcal{V}_n^{\mathrm{F\text{-}M}}(\lambda) \times \mathbb{A}_\varepsilon^{2n}$ over $\mathbb{G}_m \times \mathbb{A}^{2n}$, and the GeoVac static sub-case is the fiber over the single closed point $\{1\} \times \{0\}^{2n}$. The Grothendieck-class computation of Fathizadeh–Marcolli Lemma 7.1.5 evaluated at $\lambda = 1$ (equivalently: at the standard unit-isotropic quadric over $\mathbb{Q}(\sqrt{-1})$) gives

$$[\mathcal{V}_n^{\mathrm{GeoVac}}] = \mathbb{L}^{2n+3} - 2\mathbb{L}^{2n+2} - (\mathbb{L} - 2)(\mathbb{L} - 1)[Z_{1,2n}] - (\mathbb{L} - 2), \tag{14}$$

where $[Z_{1,2n}]$ is the Grothendieck class of the standard unit isotropic quadric in $\mathbb{P}^{2n+1}$. By the Fathizadeh–Marcolli inductive argument ([25] Theorem 7.6 and the paragraph following it), $[Z_{1,2n}]$ admits the explicit closed form

$$[Z_{1,2n}] = \mathbb{L}^{2n} + \mathbb{L}^{2n-1} + \cdots + \mathbb{L}^{n+1} + 2\mathbb{L}^n + \mathbb{L}^{n-1} + \cdots + \mathbb{L}^2 + \mathbb{L} + 1 = [\mathbb{P}^n] \cdot (1 + \mathbb{L}^n), \tag{15}$$

i.e. a palindrome of length $2n+1$ in $\mathbb{L}$ with all coefficients equal to 1 except the middle coefficient $\mathbb{L}^n$ which is doubled, factoring as $[\mathbb{P}^n] \cdot (1 + \mathbb{L}^n)$. Substituting (15) into (14) and expanding yields the fully explicit GeoVac affine-complement Grothendieck class

$$[\mathcal{V}_n^{\mathrm{GeoVac}}] = \mathbb{L}^{2n+3} - 3\mathbb{L}^{2n+2} + 2\mathbb{L}^{2n+1} - \mathbb{L}^{n+2} + 3\mathbb{L}^{n+1} - 2\mathbb{L}^n \quad \in \quad \mathbb{Z}[\mathbb{L}] \subset K_0(\mathrm{Var}_{\mathbb{Q}}), \tag{16}$$

the standard mixed-Tate Grothendieck sub-ring. At $n = 1$ the two middle terms collapse ($2n + 1 = n + 2 = 3$), giving $\mathbb{L}^5 - 3\mathbb{L}^4 + \mathbb{L}^3 + 3\mathbb{L}^2 - 2\mathbb{L}$, matching the explicit calculation in Fathizadeh–Marcolli Theorem 7.2 (verified symbolically at $n = 1, \ldots, 8$, and via the general-n symbolic identity, in `tests/test_paper55_grothendieck.py`).

Distinction from the M3 cyclotomic-level-4 ring (§5). The $\mathbb{Q}(\sqrt{-1})$ appearing here is the *Witt-splitting field* of the standard Euclidean quadratic form $Q_{1,2n}$: a generic field selected by the positive-definite signature alone, with $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$ acting *trivially* on M2 outputs (the pure-Tate ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ descends to $\mathbb{Q}$). The same field appearing in §5 as $\mathbb{Q}(\zeta_4)$ is the *cyclotomic conductor* of the non-trivial primitive Dirichlet character χ_{-4}, with $\mathrm{Gal}(\mathbb{Q}(\zeta_4)/\mathbb{Q})$ acting *non-trivially* via the Deligne–Glanois descent from level 4 to level 2. The coincidence of fields is at the number-field level only; the two motivic Galois actions are structurally independent (sprint memo `debug/sprint_a7_m2_m3_cyclotomic_memo.md`, June 2026). The factorisation $[Z_{1,2n}] = [\mathbb{P}^n] \cdot (1 + \mathbb{L}^n)$ exhibits the two split-quadric maximal isotropic subspaces over $\mathbb{Q}(i)$ as the source of the doubled middle Tate twist $\mathbb{Z}(n)[2n]$, sharpening the coincidence at the Grothendieck-class level without promoting it to a shared Tannakian symmetry; a possible deeper unification is recorded as an open question (Q5', §7).[1]

Remark 4.3 (Convention: asymptotic heat-trace vs. Vassilevich–Branson–Gilkey)**.** The closed forms in this section are stated in the volume-normalised asymptotic heat-trace convention used by Paper 51 [18] Cor. 2.1. The Vassilevich–Branson–Gilkey curvature-polynomial convention implemented by `seeley_dewitt_coefficients_s3()` in `geovac/qed_vacuum_polarization.py` returns the spinor-sector $a_0 = \sqrt{\pi}$ before $(4\pi)^{d/2}$ volume rescaling, sitting in $\sqrt{\pi} \cdot \mathbb{Q}$ at the raw level; multiplying by $\dim_S \cdot \mathrm{Vol}(S^3) = 4 \cdot 2\pi^2$ recovers the bare Dirac coefficient $8\pi^2 \in \pi^2 \cdot \mathbb{Q}$, consistent with the $a_0^{D^2} = 4\pi^2$ (vol-norm) of [18] Cor. 2.1 line 513. Both readings put the SD content in the same M2 pure-Tate ring; the classification is convention-invariant. Future readers reproducing the M2 witness panel from production code should apply the $(4\pi)^{d/2}$ rescaling to land in the asymptotic convention used here. Verified at $k = 0, 1$ in `tests/test_paper55_M2_mixed_tate.py` (2026-06-04 verification cleanup).

4.2 Witnesses

The empirical pure-Tate witness panel from the GeoVac corpus:

Witness	Source	Closed form	Tate weight
$a_0^{D^2}$ (Dirac SD, vol-norm)	Paper 51 Cor. 2.1 [18]	$4\pi^2$	$+2$
$a_1^{D^2}$ (Dirac SD, vol-norm)	Paper 51 Cor. 2.1	$-2\pi^2$	$+2$
$a_k^{D^2}$, $k \geq 2$	Paper 51 Cor. 2.1	0	—
a_k^{Δ} (scalar SD, vol-norm)	Paper 51 Thm 3.1	$2\pi^2/k!$	$+2$
$\zeta_{D^2}(2)$	Paper 28 Thm 1 [11]	$\pi^2 - \pi^4/12$	$+2, +4$
$\zeta_{D^2}(3)$	Paper 28 Thm 1	$\pi^4/3 - \pi^6/30$	$+4, +6$
$\zeta_{D^2}(4)$	Paper 28 Thm 1	$2\pi^6/15 - 17\pi^8/1260$	$+6, +8$
$\zeta_{D^2}(5)$	Paper 28 Thm 1	$\pi^8(306 - 31\pi^2)/5670$	$+8, +10$

All witnesses are in $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$; no $\sqrt{\pi}$, no $\zeta(2k+1)$, no MZV content at the SD or spectral-zeta-at-integer-s level.

4.3 Volume-normalisation convention

A structural caveat at the raw heat-trace level: the un-normalised coefficients $\tilde{a}_k = a_k/(4\pi)^{3/2}$ contain $\sqrt{\pi}$ explicitly (from the spinor fiber dimension factor that always appears in d-odd heat-kernel expansions on spin manifolds). This $\sqrt{\pi}$ cancels exactly against the $(4\pi)^{d/2}$ in the standard volume-normalisation convention, recovering the pure-Tate ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$. At the spectral-zeta-at-integer-s level (which is convention-free), the values lie in $\mathbb{Q}[\pi^2]$; the $\sqrt{\pi}$ slot is populated only at half-integer s, where the values match the leading heat-trace poles. This reconciles Paper 28 Theorem 1 (T9, $\sqrt{\pi} \cdot \mathbb{Q} \oplus \pi^2 \cdot \mathbb{Q}$ two-term ring) with the pure-Tate classification at integer s.

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

5 M3: vertex parity, Hurwitz at quarter-integer shifts, and the Deligne–Glanois Galois descent

5.1 Period-ring statement

Theorem 5.1 (M3 period-ring)**.** *The M3 sub-mechanism on the discrete Camporesi–Higuchi S^3 spectral triple produces period values in cyclotomic mixed-Tate periods at level ≤ 4 over the ring of S-integers $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$:*

$$M_3 \ \subset \ \mathcal{MT}(\mathbb{Z}[i, 1/2]), \tag{17}$$

with depth equal to loop order and three structurally distinct sub-sectors:

1. ***Un-restricted Dirac Dirichlet.*** *$D(s)$ at integer $s \geq 2$ lies in $\mathcal{MT}(\mathbb{Z}[1/2])$ at depth 1. Via the Hurwitz shift $\zeta(s, 3/2) = \zeta(s, 1/2) - 2^s$, this further reduces to $\mathcal{MT}(\mathbb{Z})$ (classical Brown MZV at level 1, [22]): $\pi^{\text{even}} \cdot \mathbb{Q}$ at even s and $\zeta(\text{odd}) \cdot \mathbb{Q}$ at odd s.*

2. ***Vertex-restricted χ_{-4} content.*** *The parity discriminant*

   $$D_{\text{even}}(s) - D_{\text{odd}}(s) \ = \ 2^{s-1}(\beta(s) - \beta(s-2))$$

   (Paper 28 Theorem 3, [11]) at integer $s \geq 2$ lies in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level $N = 4$ and depth 1. Catalan $G = \beta(2)$ and $\beta(4)$ are level-4 Dirichlet L-periods individually, accessed through Hurwitz at quarter-integer shifts $1/4, 3/4, 5/4$.

3. ***Depth-k loop tower.*** *At loop order $k \geq 2$ on the iterated sunset with vertex restriction, the M3 sector contributes an element at depth $\leq k$ in $\mathcal{MT}(\mathbb{Z}[1/2])$ at level 2. Loop order bounds the depth of the classical multiple-t-value content; it does not automatically mint new irreducibles. The $k = 2$ instance $S_{\min}$ (Paper 28 §$S_{\min}$, [11]) is structurally a depth-2 nested sum, but its* value *reduces fully: every Hoffman double t-value [30] in its decomposition is depth ≤ 1, giving $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ in closed form —* a level drop *(level 2 vs the ≤ 4 bound from sub-sector 2)* and *a depth drop. The first genuine depth-2 generators appear at $k = 3$ ($t(5, 3)$, $t(5, 1)$; three-loop decomposition program). The prior "PSLQ-irreducible" label was a basis-coverage artifact.*

The classification is a direct application of Deligne 2010 [23] and Glanois 2015 [24] to the Hurwitz / Dirichlet-L / MZV building blocks already enumerated in Paper 28.

Proof sketch. The Mellin transform at $k = 1$, $\mathcal{M}[\mathrm{Tr}(D \cdot e^{-tD^2})](s)$, extracts the spectral zeta $D(2s+1)$ via Eq. (6). On the Camporesi–Higuchi spectrum (3), the half-integer shift 3/2 writes

$$D(s) = 2\,\zeta(s-2, 3/2) - \tfrac{1}{2}\,\zeta(s, 3/2),$$

so the M3 output is a $\mathbb{Q}$-linear combination of Hurwitz zeta values at the half-integer shift 3/2. Under the elementary identity $\zeta(s, 3/2) = \zeta(s, 1/2) - 2^s$ and the duplication $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$, these reduce at integer $s \geq 2$ to $\mathbb{Q}$-linear combinations of $\zeta(s)$ and $\zeta(s-2)$, which sit in $\mathcal{MT}(\mathbb{Z})$ at depth 1 by Brown 2012 [22] (sub-sector 1). For the vertex-restricted decomposition $D = D_{\text{even}} + D_{\text{odd}}$ on the parity of the Fock quantum number n_{Fock}, the individual pieces decompose into Hurwitz at quarter-integer shifts $1/4, 3/4, 5/4$ (Paper 28 Theorem 3 proof), and the $\zeta(s, 1/4)$ contribution cancels in $D_{\text{even}} - D_{\text{odd}}$ via the Dirichlet β definition $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s\beta(s)$. Hurwitz at quarter-integer shifts are periods of $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level $N = 4$ by the explicit Glanois 2015 basis $\mathcal{B}^4$ ([24] Corollaries 1.1–1.2), so D_{even}, D_{odd}, and their difference all lie in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at depth 1 (sub-sector 2). The depth-k tower inherits its motivic content by iterated convolution: at loop order k, the iterated sunset on S^3 produces a k-fold convolution of Hurwitz building blocks, *bounding* the motivic depth by the loop order. Stuffle and parity reductions within the ring can lower the realized depth — at $k = 2$, fully to depth ≤ 1 (round-2 dossier, 2026-06-11). This is the GeoVac-side realisation of the Glanois depth-filtration $\mathrm{gr}^D_p \mathcal{H}^N$ at $N \leq 4$ (sub-sector 3). □

5.2 Arithmetic stratification

The three sub-sectors of Theorem 5.1 sit in three nested period rings selected jointly by (i) presence or absence of vertex restriction and (ii) loop order, which bounds the motivic depth:

Sub-sector	Object	Depth	Period ri
Un-restricted $D(s)$, s even	$D(2k) \in \pi^{2k} \cdot \mathbb{Q}$	1	$\mathcal{MT}(\mathbb{Z})$ pur
Un-restricted $D(s)$, s odd	$D(2k+1) \in \zeta(2k+1) \cdot \mathbb{Q}$	1	$\mathcal{MT}(\mathbb{Z})$ de
χ_{-4} vertex parity	$\beta(s) - \beta(s-2)$ via Hurwitz $1/4, 3/4$	1	$\mathcal{MT}(\mathbb{Z}[i, 1$
Catalan $G = \beta(2)$, $\beta(4)$ individually	level-4 Dirichlet L	1	$\mathcal{MT}(\mathbb{Z}[i, 1$
Sommerfeld D_p, $p \geq 2$	weight-$(2p-1)$ MZV products	≤ 2	$\mathcal{MT}(\mathbb{Z})$ Euler
$S_{\min}$ (nesting depth 2)	identified: closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$	≤ 1	$\mathcal{MT}(\mathbb{Z}[1,$
Depth-k tower, $k \geq 3$	depth-$\leq k$ element	$\leq k$	$\mathcal{MT}(\mathbb{Z}[1/2])$ a

The aggregate ring containment is

$$\mathcal{MT}(\mathbb{Z}) \subset \mathcal{MT}(\mathbb{Z}[1/2]) \subset \mathcal{MT}(\mathbb{Z}[i, 1/2]), \tag{18}$$

selected sub-sector by sub-sector by the vertex-restriction and depth data above. No M3 closed form escapes the largest of these, $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level 4.

5.3 Witness panel

5.3.1 Un-restricted $D(s)$ (Paper 28 Theorem 2)

The Dirac Dirichlet at integer $s \geq 2$ produces a clean two-pattern sequence on parity of s (Paper 28 Theorem 2 [11]):

s	$D(s)$	Class
4	$\pi^2 - \pi^4/12$	$\pi^{\text{even}} \cdot \mathbb{Q}$
5	$14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$	$\zeta(\text{odd}) \cdot \mathbb{Q}$
6	$\pi^4/3 - \pi^6/30$	$\pi^{\text{even}} \cdot \mathbb{Q}$
7	$62\,\zeta(5) - \frac{127}{2}\,\zeta(7)$	$\zeta(\text{odd}) \cdot \mathbb{Q}$
8	$2\pi^6/15 - 17\pi^8/1260$	$\pi^{\text{even}} \cdot \mathbb{Q}$
9	$254\,\zeta(7) - \frac{511}{2}\,\zeta(9)$	$\zeta(\text{odd}) \cdot \mathbb{Q}$
10	$17\pi^8/315 - 31\pi^{10}/5670$	$\pi^{\text{even}} \cdot \mathbb{Q}$

All seven entries sit in $\mathcal{MT}(\mathbb{Z})$ at depth 1.

5.3.2 χ_{-4} vertex-restricted content (Paper 28 Theorem 3)

At integer $s \geq 2$,

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\big(\beta(s) - \beta(s-2)\big).$$

Explicit values at $s = 4$:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4),$$
$$D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4),$$

where $G = \beta(2)$. Each of $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$, and their difference sits in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level 4, depth 1; their sum recovers the un-restricted $D(s)$ in the smaller level-2 ring above.

5.3.3 Sommerfeld D_p (Paper 28 §K-Sommerfeld)

Vertex-restricted Sommerfeld sums weighted by harmonic numbers $H^{(r)}_{n-1}$. Canonical closed form at $p = 4$:

$$D_4 \;=\; -\tfrac{41\pi^6}{12096} + \tfrac{71}{8}\,\zeta(7) - \tfrac{\pi^4}{20}\,\zeta(3).$$

Three-tier decomposition: π^6 (M3-π-even sub-sector), $\zeta(7)$ (M3-odd-ζ sub-sector), mixed product $\pi^4\cdot\zeta(3)$ (depth-2 MZV). At Sommerfeld order $p \geq 2$, D_p is a $\mathbb{Q}$-linear sum of weight-$(2p-1)$ MZV elements with $\max(0, \lfloor(2p-5)/2\rfloor)$ surviving cross-products $\zeta(a)\zeta(b)$, $a + b = 2p - 1$ (Paper 28 Observation product_rule). PSLQ at 80 dps against $\{G, \beta(3), \beta(4)\}$ returns no level-4 content (Paper 28 [11]); all D_p remain in $\mathcal{MT}(\mathbb{Z})$ at depth ≤ 2.

5.3.4 Two-loop identified $S_{\min}$ (Paper 28 §$S_{\min}$)

$$S_{\min} \;=\; \sum_{k=1}^{\infty} T(k)^2, \quad T(k) \;=\; 2\,\zeta(2, k+\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(4, k+\tfrac{3}{2}).$$

At 200 dps: $S_{\min} = 2.47993\,69380\,34222\,55441\,\ldots$

Correction (round-2 dossier, 2026-06-11). The 100-element PSLQ at 200 dps returned NULL — this was not irreducibility but a basis-coverage artifact: all tested bases lacked the level-2 weight-5–8 monomials ($\pi^4 \ln 2$, $\mathrm{Li}_5(1/2)$, $\mathrm{Li}_4(1/2)\ln 2$, $\zeta(3)\ln^2 2, \ldots$). Round-2 decomposition (verified to residual 1.8×10^{-129}): $S_{\min}$ decomposes into 8 Hoffman double t-values $t(b, c)$, $b \in \{2, 4\}$, $c \in \{1, 2, 3, 4\}$, with even-weight entries reduced by stuffle identities and odd-weight entries $t(2, 1)$, $t(2, 3)$ reduced to depth-1:

$$t(2,1) = -\tfrac{7}{16}\,\zeta(3) + \tfrac{1}{8}\,\pi^2 \ln 2, \quad t(2,3) = -\tfrac{31}{64}\,\zeta(5) + \tfrac{1}{16}\,\pi^2\zeta(3).$$

(Formulas match Hoffman [30] Table 2.) The remaining odd-weight values $t(4, 1)$ (weight 5) and $t(4, 3)$ (weight 7) were identified at 220 dps against the full conjectural level-2 bases (residuals $\leq 3.6 \times 10^{-222}$; explicit formulas in Paper 28 [11]). Assembling the full decomposition yields the closed form

$$S_{\min} = 8\pi^2 \ln 2 - \tfrac{2}{3}\pi^4 \ln 2 - 3\pi^2\zeta(3) + \tfrac{1}{2}\pi^4\zeta(3) - \tfrac{5}{2}\pi^2\zeta(5) + \tfrac{\pi^6}{4} - \tfrac{3}{2}\pi^4 - \tfrac{\pi^8}{96},$$

verified against the Hurwitz-tail anchor at 130 dps (residual 1.7×10^{-129}). $\zeta(7)$ cancels exactly, all rational constants cancel to zero, and every surviving term carries at least π^2. No depth-2 generator survives: $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$, realized depth ≤ 1.

Level drop. $S_{\min} \in \mathcal{MT}(\mathbb{Z}[1/2])$ at level 2 — a level drop relative to the ≤ 4 bound from sub-sector 2 of Theorem 5.1. The vertex-parity mechanism (sub-sector 2) accesses level 4 via Hurwitz at quarter-integer shifts; the min-weighted double sum $S_{\min}$ lives at level 2 via Hurwitz at half-integer shift 3/2. These are sub-sectors of the same $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ ring but are motivically distinct: level-2 content does not require the χ_{-4} descent.

5.3.5 Depth-k loop tower (Paper 28 Proposition depth_k)

Loop order k	Witness	Depth	
1	π^{even} only	1 in $\mathcal{MT}(\mathbb{Z})$	
2	$S_{\min}$: closed form	realized ≤ 1 in $\mathcal{MT}(\mathbb{Z}[1/2])$	ider
3	$S^{(3)} = 31.5726\ldots$: triple t-values	realized ≤ 2; sole depth-2 constant $\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$	fully
k general	depth-$\leq k$ element	$\leq k$ in $\mathcal{MT}(\mathbb{Z}[1/2])$	l

The empirical tower (loop order $\to$ motivic depth bound) is the GeoVac realisation of the Glanois depth filtration $\mathrm{gr}_p^D \mathcal{H}^N$ at $N = 2$. Loop order bounds the depth of the classical level-2 multiple-t-value content; stuffle and parity reductions within $\mathcal{MT}(\mathbb{Z}[1/2])$ can lower the effective depth (as demonstrated for $S_{\min}$ at $k = 2$: the structural nesting is depth 2 but *every* component reduces — stuffle identities kill the even-weight t-values and the t-value parity theorem the odd-weight ones — leaving the closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ of §5.3.4, realized depth ≤ 1; Paper 28 [11]).

5.4 The Deligne–Glanois Galois descent identification

The substantive new content of this section is the structural identification of the vertex-parity-as-χ_{-4} mechanism (the sub-sector accessing Catalan G, $\beta(4)$, and the difference $D_{\text{even}} - D_{\text{odd}}$) as the Deligne–Glanois Galois descent at $N = 4$ on the framework's natural M3 observables.

Glanois 2015 [24] establishes (Corollaries 1.1–1.2) that for $N \in \{2, 3, 4, 6, 8\}$ the period map $\mathrm{per} : \mathcal{H}^N \to \mathcal{Z}^N$ is an isomorphism, and provides an explicit basis $\mathcal{B}^N$ of motivic multiple zeta values relative to N-th roots of unity. At $N = 4$ the basis indexing data $(x_1, \ldots, x_p; \epsilon_1, \ldots, \epsilon_p)$ allows $\epsilon_p = \sqrt{-1}$ and $\epsilon_i \in \{\pm 1\}$ at other entries, with level filtration controlled by the number of even x_i entries. The descent from level 4 to level 2 is realised by restricting to the sub-basis with $\epsilon_p \in \{\pm 1\}$ (equivalently, even number of imaginary-root insertions, fixed by the Galois involution $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$).

On the GeoVac side, the parity decomposition $D = D_{\text{even}} + D_{\text{odd}}$ acts on the half-integer Camporesi–Higuchi spectrum precisely as follows: the D_{even} sector contributes Hurwitz $\zeta(s, 1/4)$ via the even-parity Fock-number selection, the D_{odd} sector contributes Hurwitz $\zeta(s, 3/4)$, and the Dirichlet β definition $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s \beta(s)$ realises the Galois involution $\chi_{-4} : n \mapsto \chi_{-4}(n)$ as the arithmetic descent operation. The cancellation $\zeta(s, 1/4) + \zeta(s, 3/4) = (4^s - 2^s)\zeta(s)$ in the sum $D_{\text{even}} + D_{\text{odd}} = D$ implements the descent from $N = 4$ to $N = 1$ in the un-restricted Dirac Dirichlet.

We summarise:

Proposition 5.2 (Vertex parity is Deligne–Glanois descent)**.** *The Galois descent from level $N = 4$ to level $N = 2$ in the Glanois 2015 cyclotomic mixed-Tate framework is realised on the discrete Camporesi–Higuchi S^3 spectral triple by the vertex-parity decomposition $D = D_{\text{even}} + D_{\text{odd}}$. The further descent from $N = 2$ to $N = 1$ (classical MZV) is realised on the un-restricted sum, which absorbs the level-4 asymmetric content via the cancellation $\zeta(s, 1/4) + \zeta(s, 3/4) = (4^s - 2^s)\zeta(s)$ at integer s.*

This identification places the parity-discriminant and χ_{-4} theorems of Paper 28 inside the standard cyclotomic-mixed-Tate Galois hierarchy at the operator-level resolution provided by the discrete spectral triple, with no new motivic conjecture required: Deligne 2010 [23] proves the period map is an isomorphism at $N = 4$, Glanois 2015 [24] provides the explicit descent operation, and Paper 28 supplies the closed forms that match the descent on the spectral-triple side.

5.5 Cross-manifold extension to S^5: same classification, three-term polynomial structure

The S^5 Bargmann–Segal Hardy lattice (Paper 24 [9]) substrate has the Camporesi–Higuchi Dirac spectrum

$$|\lambda_n^{\mathrm{CH}, S^5}| = n + \tfrac{5}{2}, \qquad g_n = \tfrac{(n+1)(n+2)(n+3)(n+4)}{3},$$

with the quartic degeneracy polynomial

$$g(m) = \tfrac{1}{3}\, m^4 - \tfrac{5}{6}\, m^2 + \tfrac{3}{16}, \qquad m = n + \tfrac{5}{2}.$$

Identities $g(1/2) = g(3/2) = 0$ extend the spectrum to all positive half-integers without spurious modes (the same Bernoulli mechanism that gives the three-term-exact S^5 scalar SD expansion in §7.3 below).

The S^5 Dirac Dirichlet series writes

$$D^{(S^5)}(s) = \tfrac{1}{3}\, Z(s-4) - \tfrac{5}{6}\, Z(s-2) + \tfrac{3}{16}\, Z(s), \tag{19}$$

where $Z(s) := \zeta(s, 1/2) - 2^s - (2/3)^s = (2^s - 1)\,\zeta(s) - 2^s - (2/3)^s$ captures the half-integer-shift-corrected sum over $m \in (\mathbb{Z} + 1/2)$, $m \geq 5/2$. This is a three-term polynomial in $\zeta(s-4), \zeta(s-2), \zeta(s)$ — one term more than the two-term S^3 counterpart, reflecting the quartic-vs-quadratic degeneracy polynomial.

Theorem 5.3 (Parity discriminant on S^5)**.** *At every even integer $s \geq 6$, $D^{(S^5)}(s)$ is purely $\pi^{\text{even}} \cdot \mathbb{Q}$ (a three-term polynomial in $\{\pi^{s-4}, \pi^{s-2}, \pi^s\}$). At every odd integer $s \geq 7$, $D^{(S^5)}(s)$ is a $\mathbb{Q}$-linear combination of $\zeta(s-4), \zeta(s-2), \zeta(s)$.*

Theorem 5.4 (χ_{-4} identity on S^5)**.** *For every integer $s \geq 6$,*

$$D^{(S^5)}_{\text{even}}(s) - D^{(S^5)}_{\text{odd}}(s) \;=\; \tfrac{1}{3}\, f_5(s-4) - \tfrac{5}{6}\, f_5(s-2) + \tfrac{3}{16}\, f_5(s), \tag{20}$$

where $f_5(s) := 2^s\, \beta(s) - 2^s + (2/3)^s$ and $\beta(s) = L(s, \chi_{-4})$.

Proof sketch. Writing $m = n + 5/2$, the parity decomposition gives $Z_{\text{even}}(s) = 2^{-s}\zeta(s, 5/4)$, $Z_{\text{odd}}(s) = 2^{-s}\zeta(s, 7/4)$. Applying the Hurwitz shift identities $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$, $\zeta(s, 7/4) = \zeta(s, 3/4) - (4/3)^s$, and the Dirichlet beta definition $\zeta(s, 1/4) - \zeta(s, 3/4) = 4^s\, \beta(s)$, the difference reduces to $f_5(s)$. Inserting into (19) yields the claimed three-term combination. The direct spectral sum, the Hurwitz closed form, and the χ_{-4} parity discriminant are cross-checked at $s \in \{7, 8, 9, 10\}$ in `tests/test_paper55_m3_s5.py` (direct-vs-closed and parity-vs-β agreement to ~ 40 digits, plus a wrong-degeneracy non-tautology guard). □

Corollary 5.5 (Cyclotomic mixed-Tate classification on S^5)**.** *$M_3^{(S^5)} \subset \mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4, depth bounded by loop order. The Glanois 2015 [24] basis $\mathcal{B}^4$ controls the period-theoretic placement; no extension to level 8 is required, because the S^5 shifts $5/4, 7/4$ reduce to the level-4 shifts $1/4, 3/4$ via the Hurwitz shift identity $\zeta(s, a+1) = \zeta(s, a) - a^{-s}$, with the $-a^{-s}$ correction adding only $\mathbb{Q}$-rational content.*

The structural transfer mechanism is identical to S^3: the vertex-parity-as-χ_{-4} Galois descent from level 4 to level 2 acts on $D^{(S^5)}$ exactly as on $D^{(S^3)}$; only the rational coefficients (and the number of polynomial terms, 3 vs. 2) differ.

Specific motivic identification of $S^{(S^5)}_{\min}$. The two-loop S^5 irreducible

$$S^{(S^5)}_{\min} := \sum_{k \geq 1} T_5(k)^2, \quad T_5(k) := \tfrac{1}{3}\zeta(2, k + \tfrac{5}{2}) - \tfrac{5}{6}\zeta(4, k + \tfrac{5}{2}) + \tfrac{3}{16}\zeta(6, k + \tfrac{5}{2}),$$

truncates to $S^{(S^5)}_{\min} \approx 0.0399612091165\ldots$ at $K_{\max} = 500$, 50 dps. The natural verification target is the decomposition-first pipeline that identified the S^3 case (round-2 dossier, 2026-06-11): weight-grade the object, expand into Hoffman t-values at half-integer shift, apply stuffle and parity reductions, and only then run weight-homogeneous level-2 PSLQ on any surviving components. (Grab-bag single-weight PSLQ against the original Paper 28 basis is exactly the design that produced the false S^3 "irreducibility" verdict.) The natural prediction is $S^{(S^5)}_{\min} \in \mathcal{MT}(\mathbb{Z}[1/2])$ as an identifiable classical element, with nesting depth 2 and parity reductions plausibly collapsing the realized depth, as at S^3.

5.6 Diagonal-substrate collapse of the depth-2 joint Mellin

The Sprint NA-1 depth-2 Mellin test (`debug/sprint_na1_depth2_mellin_memo.md`, 2026-06-06) returned a structural identity that refines the M3 picture on the natural Camporesi–Higuchi diagonal substrate.

Theorem 5.6 (NA-1 diagonal-collapse identity)**.** *Let D be the Camporesi–Higuchi Dirac on S^3 and $\gamma_P|n,\chi,m_l\rangle := (-1)^n|n,\chi,m_l\rangle$ the vertex-parity grading on the eigenshells. For every integer $s_1+s_2 \geq 3$,*

$$J(s_1,s_2) \;:=\; \frac{1}{\Gamma(s_1)\Gamma(s_2)}\int_0^\infty\!\!\int_0^\infty t_1^{s_1-1}t_2^{s_2-1}\,\mathrm{Tr}\big(D^2\,e^{-t_1D^2}\cdot\gamma_P\,D\,e^{-t_2D^2}\big)\,dt_1\,dt_2 \tag{21}$$

depends only on $s_{\rm tot} := s_1+s_2$ and collapses bit-exactly to a depth-1 parity-graded Mellin moment at shifted argument:

$$J(s_1,s_2) \;=\; M_3^{\gamma_P}(s_{\rm tot}-1), \qquad M_3^{\gamma_P}(s) \;:=\; \sum_{n\geq 0}(-1)^n\,(n+1)(n+2)\,\big((2n+3)/2\big)^{1-2s}. \tag{22}$$

Proof sketch. On the diagonal Camporesi–Higuchi eigenbasis, $D|n\rangle = \lambda_n|n\rangle$, $\gamma_P|n\rangle = (-1)^n|n\rangle$, and $e^{-tD^2}|n\rangle = e^{-t\lambda_n^2}|n\rangle$ are simultaneously diagonal; the joint operator trace algebraically degenerates to a single sum $\sum_n(-1)^n g_n\,\lambda_n^3\,e^{-(t_1+t_2)\lambda_n^2}$. Mellin-double-transforming with respect to t_1,t_2 collects the two exponents into $s_{\rm tot} = s_1+s_2$, yielding $J(s_1,s_2) = \sum_n(-1)^n g_n\,\lambda_n^{3-2s_{\rm tot}}$. Setting $|\lambda_n| = (2n+3)/2$ and $g_n = (n+1)(n+2)$ and identifying with $M_3^{\gamma_P}(s_{\rm tot}-1)$ gives the claimed termwise algebraic identity. Bit-exact numerical verification at $s_{\rm tot} \in \{3,4,5,6,7\}$, 200 dps, recorded in `debug/data/na1_depth2_mellin_results.json`. □

The depth-2 joint Mellin transform of an M2 × M3 pair, on the natural diagonal substrate, is thus a depth-1 object in disguise. The factorisation diagnostic $J(s_1,s_2) \stackrel{?}{=} M_2(s_1)M_3(s_2)$ (primitive product) vs. $J(s_1,s_2) \stackrel{?}{=} M_2(s_1)M_3(s_2) + M_3(s_1)M_2(s_2)$ (deconcatenation pair) is *structurally invisible* on this substrate because operator ordering is trivial.

Theorem 5.7 (Closed form for $M_3^{\gamma_P}$)**.** *For every integer $s \geq 2$,*

$$M_3^{\gamma_P}(s) \;=\; 2^{2s-3}\big(\beta(2s-1)-\beta(2s-3)\big), \tag{23}$$

where $\beta(s) = L(s,\chi_{-4})$ is the Dirichlet beta function. In particular, $M_3^{\gamma_P}(s) \in \pi^{\rm odd}\cdot\mathbb{Q}$ for every integer s: the natural diagonal-substrate parity-graded Mellin moment is pure-Tate *(level 1), via the Euler-number closed form $\beta(2k+1) \in \pi^{2k+1}\cdot\mathbb{Q}$.*

Proof sketch. Substituting $u = n+3/2$ identifies the sum with $\sum_{u\in\mathbb{Z}+1/2,\,u\geq 3/2}(-1)^{u-3/2}(u^{3-2s} - \frac{1}{4}\,u^{1-2s})$. Applying the alternating-shift identity $\sum_{n\geq 0}(-1)^n(n+3/2)^{-k} = 2^k(1-\beta(k))$ to both terms and simplifying gives (23). Verified bit-exact at $s \in \{2,3,4,5,6\}$ via PSLQ at 50, 100, 200 dps in `debug/sprint_na1_depth2_mellin_compute.py`. □

Remark 5.8 (Relation to the level-4 cyclotomic content of Section 5.4)**.** Theorem 5.7 samples the parity-graded Mellin function $\sum_n(-1)^n g_n|\lambda_n|^{-r}$ at $r = 2s-1 \in 2\mathbb{Z}+1$ (odd integer), which lands inside $\pi^{\rm odd}\cdot\mathbb{Q}$ via $\beta(\text{odd}) \in \pi^{\rm odd}\cdot\mathbb{Q}$. The vertex-parity decomposition of Proposition 5.2, in contrast, probes $D_{\rm even}(s) - D_{\rm odd}(s)$ at integer s via the unshifted Dirac Dirichlet series, which accesses $\beta(\text{even})$ values (Catalan $G = \beta(2)$, $\beta(4)$) realising level-4 cyclotomic content. Both probes act on the same diagonal parity grading γ_P; the period content (pure-Tate vs level-4 cyclotomic) is selected by the Mellin slot exponent, not by a different operator. The Glanois [24] basis $\mathcal{B}^4$ accommodates both columns: the pure-Tate column is the $N = 1$ sub-basis, the level-4 column is the genuinely $N = 4$ content.

Remark 5.9 (Reading A vs Reading B on a diagonal substrate)**.** Theorem 5.6 initially suggested that the A-vs-B disambiguation question (whether the GeoVac substrate respects primitive product or shuffle deconcatenation at depth 2) is *structurally invisible* on the natural diagonal Camporesi–Higuchi substrate but might be exposed by a non-diagonal substrate in which γ_P does not commute with the operator under the Mellin slot. Theorem 5.10 below *deflates* this hope: the depth-2 trace collapse holds on *any* substrate, commuting or non-commuting, because the trace functional is blind to off-diagonal entries of γ in the D-eigenbasis. Substrate enrichment cannot recover the A-vs-B distinction at the trace-of-single-γ-insertion level.

Theorem 5.10 (NA-1-offdiag trace-functional collapse, Reading C-strong)**.** *Let D be a self-adjoint operator on a finite-dimensional Hilbert space $\mathcal{H}$ and γ* any *self-adjoint operator on $\mathcal{H}$ — not necessarily commuting with D. Let $\{\lambda_i\}_{i\in I}$ be the eigenvalues of D and $\widetilde{\gamma} := U^*\gamma U$ the matrix of γ in the eigenbasis of D. Then for every $s_1, s_2 \in \mathbb{C}$ with $\Re(s_1) + \Re(s_2) > 0$ and $s_1 + s_2 - 1$ outside the D-spectral support,*

$$J(s_1, s_2) \;:=\; \frac{1}{\Gamma(s_1)\Gamma(s_2)} \int_0^\infty\!\!\int_0^\infty t_1^{s_1-1} t_2^{s_2-1}\, \mathrm{Tr}\big(D^2\, e^{-t_1 D^2}\, \gamma\, D\, e^{-t_2 D^2}\big)\, dt_1\, dt_2 \tag{24}$$

depends only on $s_{\mathrm{tot}} := s_1 + s_2$, with closed form

$$J(s_1, s_2) \;=\; \sum_{i\in I} \widetilde{\gamma}_{ii}\, \lambda_i^{3-2s_{\mathrm{tot}}}. \tag{25}$$

Only the diagonal *entries of $\widetilde{\gamma}$ in the D-eigenbasis enter; the off-diagonal entries $\widetilde{\gamma}_{ij}$ for $i \neq j$ are invisible to the trace.*

Proof. In the eigenbasis of D, the operators $D^k e^{-tD^2}$ are diagonal with i-th entry $\lambda_i^k e^{-t\lambda_i^2}$. For diagonal operators A, B with entries A_{ii}, B_{ii} and an arbitrary operator γ, $\mathrm{Tr}(A\gamma B) = \sum_i A_{ii}\, \widetilde{\gamma}_{ii}\, B_{ii}$ because cross-terms $A_{ii}\, \widetilde{\gamma}_{ij}\, B_{jj}$ for $i \neq j$ are off-diagonal in $A\gamma B$ and don't contribute to the trace. Substituting $A = D^2 e^{-t_1 D^2}$, $B = De^{-t_2 D^2}$ gives $\mathrm{Tr}(D^2 e^{-t_1 D^2}\gamma D e^{-t_2 D^2}) = \sum_i \widetilde{\gamma}_{ii}\, \lambda_i^3\, e^{-(t_1+t_2)\lambda_i^2}$, which depends only on $t_1 + t_2$. Mellin-transforming yields $J(s_1, s_2) = \sum_i \widetilde{\gamma}_{ii}\, \lambda_i^{3-2s_{\mathrm{tot}}}$ as claimed. □

Theorem 5.6 is the specialisation to $D = D^{\mathrm{CH}}_{S^3}$ and $\gamma = \gamma_P$ vertex parity, where $\widetilde{\gamma}_P$ is itself diagonal with entries $(-1)^n$ and the sum reduces to $M_3^{\gamma_P}(s_{\mathrm{tot}} - 1)$ with the Euler-number closed form $M_3^{\gamma_P}(s) = 2^{2s-3}(\beta(2s-1) - \beta(2s-3))$ of Theorem 5.7.

Remark 5.11 (Reading A vs Reading B requires a non-trace probe)**.** Theorem 5.10 sharpens the honest statement of the open Reading A/B question. The disambiguation cannot be tested by any chain $\mathrm{Tr}(\cdots e^{-t_i D^2}\, \gamma_i\, e^{-t_j D^2} \cdots)$ with single-operator insertions between heat-kernel factors, on any substrate: the trace functional commits to the diagonal of the inserted operator in the D-eigenbasis and loses the off-diagonal information that would carry primitive-vs-shuffle content. The genuine A/B probe lives at the *Jaffe–Lesniewski–Osterwalder* entire cyclic cocycle level (the depth-2 specialisation $\chi_D(a_0, a_1, a_2)$ with two non-D-functions between heat-kernels), where the cocycle structure carries the operator ordering algebraically rather than projecting onto the trace diagonal. See `debug/sprint_na1_offdiag_substrate_memo.md` (2026-06-06) for the substrate audit at $n_{\max} \in \{2, 3\}$ on the off-diagonal CH with chirality-flipping E_1 entries, which verifies the trace-functional collapse bit-exactly even in the genuinely non-commuting case.

Theorem 5.12 (JLO depth-2 cocommutativity — Reading A empirical close)**.** *Let $\chi_D(a_0, a_1, a_2; t) = \sum_{m\geq 0} t^m\, c_m(a_0, a_1, a_2)$ be the JLO depth-2 entire cyclic cocycle on the truncated Camporesi–Higuchi spectral triple at $n_{\max} \in \{2, 3\}$, with the sector-idempotent algebra $\mathcal{A} = \mathbb{C}\langle e_0, \ldots, e_{N-1}\rangle$ ($N = N(n_{\max}) = 5, 9$ respectively). Then $\chi_D(a_0, a_1, a_2; t)$ is bit-exactly invariant under the full S_3 permutation action on (a_0, a_1, a_2) at both leading order (t^0) and the next order (t^1), in both* *the even (chirality-graded) and odd (plain trace) flavours. Concretely, on the* 25*-pair* $\{1, e_s, e_t\}$ *panel at* $n_{\max} = 2$*, the swap-asymmetric residual* $\chi_D(1, e_s, e_t) - \chi_D(1, e_t, e_s)$ *vanishes identically in* `sympy`*-rational arithmetic across all* 10 *unordered pairs* $\times$ 2 *orders* $\times$ 2 *flavours* $= 40$ *bit-exact zeros; the full* S_3*-orbit test on* $\chi_D(e_0, e_1, e_2)$ *returns the same value across all six permutations. The depth-2 content sits entirely in the* symmetric *sector of* $S_3 \curvearrowright \mathcal{A}^{\otimes 3}$*, with zero antisymmetric / shuffle-Hopf content.*

Proof sketch and honest scope. The cocycle values are computed bit-exactly via the t-power-series expansion of the JLO simplex integral (driver `debug/sprint_jlo_depth2_compute.py`). Bit-exact swap symmetry on the 10-pair $\times$ 2-order $\times$ 2-flavour panel = 40 zero residuals computed in `sympy` rational arithmetic; bit-exact S_3-orbit invariance on $\chi_D(e_0, e_1, e_2)$ across all 6 permutations.

The honest scope caveat: since $\mathcal{A} = \mathbb{C}\langle e_0, \ldots, e_{N-1}\rangle$ is commutative ($[e_s, e_t] = 0$), trace cyclicity plus commutativity of $\mathcal{A}$ alone would force *cyclic invariance modulo explicit coboundary terms*, which is weaker than the bit-exact full S_3-cocommutativity verified here. The operator-level commutators that would have to vanish to reduce S_3 invariance to cyclicity (e.g. $[\gamma, [D, e_s]]$ and $[[D, e_s], [D, e_t]]$) are *non-zero* on this substrate (e.g. $\|[\gamma, D]\| = \sqrt{13}/4$, $\|[[D, e_0], [D, e_1]]\| = 3/64$ at $n_{\max} = 2$), so the S_3-invariance is strictly stronger than what commutativity of $\mathcal{A}$ or cyclicity of the trace alone would give. See `debug/sprint_jlo_depth2_memo.md` §4.3 for the explicit non-vanishing operator commutators.

The verdict is stable across cutoff: spot-checks at $n_{\max} = 3$ reproduce bit-exact swap symmetry and bit-exact S_3-orbit invariance on the tested arguments. □

Remark 5.13 (Strategic implication for the cosmic-Galois positioning)**.** Theorem 5.12 closes the depth-2 Reading A/B question on the natural Camporesi–Higuchi substrate at the JLO cocycle level. GeoVac IS the abelianisation-by-Cartier–Milnor–Moore of the cosmic-Galois universe at depth 2: the depth-2 cocycle carries genuinely new content not derivable from depth-1 (depth-1 JLO vanishes identically on commutative $\mathcal{A}$, v3.59.0 Sprint Q5'-Stage1-Arc), but that new content sits entirely in the symmetric / primitive-product sector of $S_3 \curvearrowright \mathcal{A}^{\otimes 3}$, with zero shuffle-Hopf / deconcatenation-pair signature.

No empirical demand from the JLO probe for the beyond-sprint-scale follow-on to cofree-cocommutative shuffle Hopf algebra $T(V)$ enrichment of the substrate. The injection theorem $U^*_{\mathrm{GV}} \hookrightarrow \mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ of Paper 56 (Theorem `thm:injection_g4`, 2026-06-06) is the structurally correct statement: GeoVac is the abelianised image, not a partial achievement.

Two open questions remain. (i) Non-commutative inner-factor enrichment of $\mathcal{A}$ would expose Reading B content sitting at higher Mellin slot indices; but Sprint H1's Yukawa-non-selection theorem (Paper 32 [13] § VIII.C; `memory/sprint_h1_positive_thin.md`) already classifies the inner factor as calibration data, not autonomously generated by the GeoVac outer triple, so the non-commutative enrichment question is not a sprint-scale follow-on. (ii) The equality direction Φ^{inj} surjective onto $\mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ (Paper 56, 2026-06-06) remains the named open forward research question, testable by exhausting GeoVac's M3 sector at depth ≥ 2 for level-4 cyclotomic motivic MZVs. Today's verdict eliminates one structural ambiguity (primitive vs shuffle) on the GeoVac side without advancing the equality direction itself.

6 Joint engagement and physical observables

6.1 Role-disjointness vs set-theoretic overlap

The three sub-mechanisms M1, M2, M3 are role-disjoint at the Mellin slot index $k \in \{0, 1, 2\}$: each *elementary spectral object* of GeoVac is produced by exactly one Mellin slot of $\mathrm{Tr}(D^k e^{-tD^2})$ at one operator order k. This is a structural property of the master Mellin engine, inherited from the case-exhaustion theorem (Theorem 2.1).

The corresponding period rings M_1, M_2, M_3, however, are not set-theoretically disjoint. The nesting structure is

$$M_2 = \bigoplus_{k\geq 0} \pi^{2k} \cdot \mathbb{Q} \ \subset\ \mathcal{MT}(\mathbb{Z}) \ \subset\ M_3 = \mathcal{MT}(\mathbb{Z}[i, 1/2]), \qquad M_1 \cap \mathcal{MT}(\mathbb{Z}[i, 1/2]) = \mathbb{Q}[\pi]. \quad (26)$$

A given period value π^{2k} can be reached by M1 (as an integer power of the Hopf-base measure), by M2 (as a Seeley–DeWitt coefficient at order k), or by M3 (as an even-weight Brown MZV inside the un-restricted Dirac Dirichlet). The Mellin slot index k is the structural label that distinguishes the three readings; the period value alone does not.

The rational subring $\mathbb{Q}$ sits inside every M_i at Tate weight 0 and depth 0; rational coefficients (the Casimir trace $B = 42$ of Paper 2, the Dirac mode count $\Delta = 1/40$, $\mathbb{Q}$-linear combination weights throughout) are mechanism-tagged by what produces them, not by their numerical value. This is the cleanest form of the role-disjointness statement: *mechanisms* are disjoint; *values* can overlap.

6.2 Tate-weight bookkeeping

For a physical observable that decomposes as a product of M1, M2, M3 contributions

$$\mathcal{O} = \alpha_1 \cdot \alpha_2 \cdot \alpha_3, \quad \alpha_i \in M_i, \quad (27)$$

the Tate weight is additive,

$$\mathrm{wt}(\mathcal{O}) = \mathrm{wt}(\alpha_1) + \mathrm{wt}(\alpha_2) + \mathrm{wt}(\alpha_3), \quad (28)$$

and the motivic depth is sub-additive,

$$\mathrm{dep}(\mathcal{O}) \leq \mathrm{dep}(\alpha_1) + \mathrm{dep}(\alpha_2) + \mathrm{dep}(\alpha_3), \quad (29)$$

with equality when the factors are algebraically independent (the generic case). For an observable that decomposes as an additive combination $\mathcal{O} = \alpha_1 + \alpha_2 + \alpha_3$, the Tate weight is the maximum of the constituent weights (the weight grading is preserved by sums only within a fixed weight component), and motivic depth is again sub-additive. Joint-engagement bookkeeping for a general physical observable $\mathcal{O}$ thus follows the rule: decompose into a sum of products $\mathcal{O} = \sum_j \prod_i \alpha_{i,j}$, then apply (28)–(29) to each summand and take the maximum over j.

6.3 Canonical example 1: the K-formula $K = \pi(B + F - \Delta)$

The Paper 2 [6] combination rule

$$K = \pi\,(B + F - \Delta), \qquad B = 42, \ \ F = \zeta(2) = \tfrac{\pi^2}{6}, \ \ \Delta = \tfrac{1}{40}, \quad (30)$$

contains exactly one elementary spectral object from each of the three Mellin slots, plus a rational scalar:

- The explicit prefactor π is the Hopf-base measure from M1, Tate weight $+1$, depth 0.
- $B = 42$ is a rational Casimir trace at the $k = 0$ scalar sub-mechanism (the rational core of M1), Tate weight 0, depth 0.
- $F = \zeta(2) = \pi^2/6$ is the Fock-Dirichlet evaluation at $s = 2$ on the scalar Laplacian, accessed via the M2 Mellin slot at $k = 2$ (Seeley–DeWitt at integer s; cf. Paper 28 Theorem 1, [11]), Tate weight $+2$, depth 0.
- $\Delta = 1/40$ is the inverse g_3^{Dirac} Dirac mode degeneracy at $n = 3$, accessed via the M3 Mellin slot at $k = 1$ as a rational vertex-counting coefficient, Tate weight 0, depth 0.

Multiplying out:

$$K \;=\; \pi B - \pi\Delta + \pi F \;=\; \underbrace{\pi(B-\Delta)}_{\text{weight }+1} + \underbrace{\pi^3/6}_{\text{weight }+3}\,.$$

The constant K sits in $\mathbb{Q}[\pi]$ at weight ≤ 3, in $M_1 \cdot (M_0 \oplus M_2)$. This is the canonical *additive* joint engagement: one representative per Mellin slot summed inside a single linear combination, with the dominant term $\pi^3/6$ recognisable as the M1×M2 cross-product of Hopf-base measure and scalar Seeley–DeWitt evaluation at $s = 2$. The empirical equality $K = \alpha^{-1}$ to 8.8×10^{-8} relative precision is the Paper 2 observation; the present paper does not derive it, but places each term inside a named motivic home.

6.4 Canonical example 2: Paper 50 F-theorem coefficients

Paper 50 [17] computes the boundary-CFT$_3$ F-theorem coefficients on the truncated S^3 spectral triple for the conformally-coupled scalar and the free Camporesi–Higuchi Dirac:

$$F_s = \frac{\log 2}{8} - \frac{3\,\zeta(3)}{16\pi^2}, \tag{31}$$

$$F_D = \frac{\log 2}{4} + \frac{3\,\zeta(3)}{8\pi^2}. \tag{32}$$

Both observables are non-trivial multiplicative joint engagements between M1 and M3:

- $\log 2$ is M3 at level $N = 2$, depth 1 (a Glanois $\mathcal{B}^2$ basis element); Tate weight $+1$ in the cyclotomic grading.
- $\zeta(3)/\pi^2$ is the canonical M1×M3 cross-product: $\zeta(3) \in M_3 \cap \mathcal{MT}(\mathbb{Z})$ at Tate weight $+3$ depth 1, and $1/\pi^2 \in M_1$ at Tate weight -2 depth 0. Combined Tate weight $+1$, matching $\log 2$.

The match in Tate weight between the two summands of F_s and F_D is the internal consistency of a weight-1 period; if the cancellations required a weight mismatch, the F-theorem coefficient could not be single-valued in the cyclotomic-mixed-Tate framework.

Paper 50 Theorem 3.5 isolates the M3 and M1×M3 sectors via the orthogonal projection

$$F_D + 2F_s = \frac{\log 2}{2} \quad \text{(isolates M3-only sub-sector; all } \zeta(3) \text{ cancels)}, \tag{33}$$

$$F_D - 2F_s = \frac{3\,\zeta(3)}{4\pi^2} \quad \text{(isolates M1×M3 cross-product; all } \log 2 \text{ cancels)}. \tag{34}$$

The dual-basis projection in (33)–(34) is the prototypical experimental form of joint-engagement bookkeeping: by linearly combining two physical observables (F_s and F_D), the contribution of one mechanism (or cross-product of mechanisms) can be isolated empirically. Each side of the projection is a single-mechanism observable in a named period ring of Sections 3–5.

6.5 Canonical example 3: Paper 38 universal state-space GH rate

The asymptotic rate constant $4/\pi$ in the van Suijlekom state-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples (Paper 38 Theorem L2, [14]; universal across compact connected Lie groups, Paper 40 Theorem 1, [15]) is a pure M1 expression:

$$\frac{4}{\pi} \;=\; \frac{\mathrm{Vol}(S^2)}{\pi^2}. \tag{35}$$

The numerator $\mathrm{Vol}(S^2) = 4\pi$ is the canonical M1 Hopf-base measure (Tate weight $+1$); the denominator π^2 is the squared inverse period (Tate weight -2). Net Tate weight -1, depth 0.

No M2 or M3 content enters at any step of the state-space GH convergence argument. This is the operator-algebraic statement of the fact that state-space GH convergence rates measure the Hopf-base-measure content of the substrate independent of its heat-kernel and vertex-parity sectors: the rate constant lives in M_1 alone, and the universality across compact connected Lie groups (Paper 40) [15] is the universality of the Hopf-base-measure construction.

6.6 Joint-engagement modes

The three examples above exhibit the three natural composition modes:

Mode	Combination	Example	Ring
Additive	$\sum_i \alpha_i$ inside a single linear combination	Paper 2 K-formula	$M_1 \cdot (M_0 \oplus M_2 \oplus M$
Multiplicative	$\prod_i \alpha_i$ across mechanism boundaries	Paper 50 F-theorem	$M_1 \cdot M_3 \subset \mathcal{MT}(\mathbb{Z}[i$
Pure single-mechanism	all factors from one slot	Paper 38 rate	M_1 alone

The dual-basis projection of Paper 50 Theorem 3.5 demonstrates the operational utility of joint-engagement bookkeeping: linear combinations of physical observables that isolate one mechanism (or one cross-product) at a time make the projective content of each sub-mechanism readable from empirical data. This is the operational form of the periods-of-GeoVac classification — every physical observable decomposes into a sum of products of M1, M2, M3 factors, each factor sits in a named period ring of Sections 3–5, and the dual-basis projections recover the contribution of each Mellin slot independently.

7 Open questions

7.1 Specific motivic identification of the depth-k tower

The depth-k loop tower (Paper 28 Proposition depth_k, corrected, [11]): loop order *bounds* the depth of the classical level-2 multiple-t-value content; it does not mint new irreducible transcendentals automatically. Status by rung:

- $k = 1$: π^{even} only (Paper 28 Theorem T9, proven; depth 1 in $\mathcal{MT}(\mathbb{Z})$).
- $k = 2$: **Resolved (2026-06-11).** $S_{\min}$ is identified in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (§5.3.4); realized depth ≤ 1. The prior "PSLQ-irreducible at 200 dps" verdict was a basis-coverage artifact: the tested bases contained none of the level-2 weight-5–8 monomials the correct room requires.
- $k = 3$: **Closed (2026-06-11, closure sprint).** Both named open items resolved: the trailing-argument-1 evaluator was repaired (Abel summation; the stage-1 Levin values were mis-converged with precision-dependent error, poisoning the earlier figure $S^{(3)} \approx 30.615$), and the global anchor closed by rigorous bracket, $S^{(3)} \in [31.57063, 31.57300]$, with canonical value $S^{(3)} = 31.5726\ldots$ inside. All four pending components identified (residuals $\leq 2.6 \times 10^{-216}$): every identified component of $S^{(3)}$ — all but the weight-10 graded piece — has realized depth ≤ 2, with the catalogued generators $t(5,1)$, $t(5,3)$, $t(7,1)$ at coefficients $\pm\frac{1}{2}$ (Paper 28 §Decomposition closure; falsifier `tests/test_s3_decomposition.py`). The earlier "$S^{(3)}$ independent of $S_{\min}$" statement (20-digit PSLQ at non-physical exponents) is superseded. Same-day follow-on sprints sharpened the verdict twice: in the *full assembly* every depth-2 generator cancels exactly (the identified part of $S^{(3)}$ is a polynomial in π^2, $\ln 2$, odd zetas — with $\zeta(11)$ cancelling, the echo of $S_{\min}$'s $\zeta(7)$ cancellation), and the weight-10 block reduces by exact stuffle closure to five surviving generators

(three antisymmetric doubles + two parity-reducible triples; Paper 28 eq. “W_{10} reduction”). **W10 closed (2026-06-12):** the Charlton–Hoffman symmetry theorem (explicit-product form, Thm. 2.21 of arXiv:2204.14183) plus four clean stuffles identify all four weight-10 triples individually; in the full assembly the three antisymmetric doubles cancel identically, and $S^{(3)} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5,3)]$ explicitly — the sole depth-2 constant is the *level-1* generator $\zeta(5,3)$, at coefficient $6\pi^2$ (assembly gate 1.2×10^{-198}; falsifier `tests/test_s3_w10_identification.py`). New depth arrives from level 1; the level-2 sector stays at realized depth ≤ 1. All weight-≤ 7 closed forms are verbatim rows of Hoffman's published Appendix A — the PSLQ identifications independently rediscovered the table, and the $t_3(3,4,1)$ explicit form appears to be new.

- k general (open): does the realized depth keep growing strictly more slowly than loop order, and does a genuinely irreducible (but catalogued) level-2 depth-k t-value survive at some k? Realized depth $\leq k-1$ at $k = 2, 3$; at $k = 3$ no level-2 generator survives at all (the depth-2 content is the level-1 $\zeta(5,3)$), so the sharper open question is whether *any* level-2-specific constant beyond the depth-≤ 1 ring ever survives a full chain assembly. Both previously named remaining items (the weight-10 graded room and the full symbolic assembly) are closed (W10 sprint, 2026-06-12).

Method discipline (round-2 dossier lesson): weight-grade the object first, then identify each graded component against a complete weight-homogeneous level-2 basis. Grab-bag single-weight PSLQ produces false irreducibility verdicts on weight-inhomogeneous objects.

7.2 Inhomogeneous extension: three sub-cases

The Fathizadeh–Marcolli result holds on $\mathbb{R} \times S^3$ with arbitrary scaling factor $a(t)$; the present paper restricts to $a(t) \equiv 1$, the closed-point specialisation $\{1\} \times \{0\}^{2n}$ of §4. Scoping (Sprint A1, 2026-06-03; memo `debug/sprint_a1_scoping_memo.md`) splits “inhomogeneous extension” into three structurally distinct sub-cases:

- *Constant warp.* Rescaling to $a(t) \equiv a_0 > 0$ rational corresponds to the λ-slice $\{\lambda = a_0\} \times \{0\}^{2n} \hookrightarrow \mathbb{G}_m \times \mathbb{A}^{2n}$. The pure-Tate ring is preserved by rational rescaling because Paper 51 [18] Theorem `thm:zeta_unit_neg_k` ($\zeta_{S^3,R}(-k) = 0$ for all k) gives a two-term exact spectral action at all R. No sprint required; trivial corollary of §4.
- *Thermal-time compactification (closed in the negative).* The temporal compactification $\mathbb{R} \to S^1_\beta$ (the natural GeoVac time-direction surrogate via the Paper 35 [12] Matsubara substrate $S^3 \times S^1_\beta$, not literally an $a(t)$ in F–M's sense) does NOT inject classical odd-zeta $\zeta(2k+1)$ or MZV content into the 4D Seeley–DeWitt coefficients. Sprint A1-Matsubara (2026-06-03, `debug/sprint_a1_matsubara_memo.md`) computes the small-t asymptotic of the product heat kernel $K_{S^3 \times S^1_\beta}(t) = K_{S^3}(t) \cdot K_{S^1_\beta}(t)$ and finds that the temporal factor contributes a single power $\beta/\sqrt{4\pi t}$ at $t \to 0^+$ (the $m = 0$ winding of ϑ_3), with the $m \neq 0$ winding terms exponentially suppressed by $e^{-m^2\beta^2/(4t)}$. The Matsubara mode sum — the mechanism by which $\zeta(2k+1)$ could enter — lives in the orthogonal large-t (IR) regime of $K_{S^1_\beta}(t)$, reached only via Jacobi ϑ_3 modular inversion, and is not visible to the SD asymptotic.

  Proposition 7.1 (Pure-Tate refinement on $S^3 \times S^1_\beta$)**.** *The volume-normalised 4D Seeley–DeWitt coefficients on $S^3 \times S^1_\beta$ for the scalar Laplacian and the Camporesi–Higuchi Dirac satisfy*

$$a_k^{\text{4D,scalar}}(\beta) = \frac{2\pi^2\,\beta}{k!} \quad (k \geq 0), \tag{36}$$

$$a_0^{\text{4D,Dirac}}(\beta) = 4\pi^2\,\beta, \quad a_1^{\text{4D,Dirac}}(\beta) = -2\pi^2\,\beta, \quad a_k^{\text{4D,Dirac}}(\beta) = 0 \;\; (k \geq 2). \tag{37}$$

Each coefficient factorises as $\beta \cdot a_k^{S^3}$, where $a_k^{S^3}$ is the corresponding SD coefficient on the static S^3 sub-case (§4, Paper 51 Thm 3.1 and Cor. 2.1 [18]). The two-term exactness of the S^3 Dirac sector is inherited verbatim. All coefficients therefore lie in the rational-rescaled pure-Tate ring

$$M_2^{(\mathrm{4D,thermal})} \subset \beta \cdot \bigoplus_{k\geq 0} \pi^{2k} \cdot \mathbb{Q}.$$

The Stefan–Boltzmann π-injection of Paper 35 § III is structurally a different extraction — the Mellin transform at integer s on the same product heat kernel, not the small-t SD asymptotic — and lives in the M1×M2 cross-product joint engagement of §6, not in M2 itself. No conflict.

- *Generic continuum $a(t)$ with non-trivial derivatives $\varepsilon_i = a^{(i)}(t) \neq 0$.* Requires either a 4D continuum substrate or a time-dependent Camporesi–Higuchi spectral triple, both outside the discrete-substrate principle that defines GeoVac. The closest GeoVac analog is the discrete-warped-substrate program of Paper 51 [18] §G4-3, currently a follow-on beyond sprint scale. No active sprint.

The original “inhomogeneous extension” open question is therefore stratified, with constant warp closed by Paper 51 §G1, A1-Matsubara named as a new sprint-scale open question, and the full generic case recorded as a deep structural frontier tied to the Paper 51 discrete-warped-substrate program.

7.3 Cross-manifold extension to S^5 Bargmann–Segal: M2 closes by dimension-parity sharpening

The M2 sub-mechanism on the Hardy sector of S^5 (Paper 24 substrate [9]) inherits the F–M mixed-Tate classification verbatim, refined by the dimension parity of the substrate. On round unit S^5:

Proposition 7.2 (Pure-Tate refinement on S^5)**.** *The volume-normalised Seeley–DeWitt coefficients of the Camporesi–Higuchi Dirac D^2 and the scalar Laplacian on round unit S^5 are:*

- *Dirac (three-term exact). Exactly three non-zero coefficients,*

$$a_0^{D^2} = 4\pi^3, \qquad a_1^{D^2} = -\tfrac{20}{3}\pi^3, \qquad a_2^{D^2} = 3\pi^3,$$

  and $a_k^{D^2} = 0$ for $k \geq 3$. The three-term truncation matches the general S^{2m+1} count $(m+1)$ at $m = 2$ (Paper 51 `rem:two_term_uniqueness`, [18]).

- *Scalar Laplacian (infinite closed form).*

$$a_0^{\Delta} = \pi^3, \qquad a_k^{\Delta} = \frac{(6-k)\cdot 4^{k-1}\cdot 2}{3\cdot k!}\,\pi^3 \quad (k \geq 1),$$

  with the single mid-series zero $a_6^{\Delta} = 0$ from the $(6-k)$ factor.

All coefficients lie in the single Tate-weight-3 slice $\pi^3 \cdot \mathbb{Q} \subset \mathbb{Q}[\pi] \subset \mathcal{MT}(\mathbb{Q})$. Consequently

$$M_2^{(S^5)} \ \subset\ \pi^3 \cdot \mathbb{Q}, \tag{38}$$

which sits in the same Kontsevich–Zagier mixed-Tate-over-$\mathbb{Q}$ class as the S^3 counterpart but in a strictly different single-weight slice (odd weight +3 rather than even-weight sub-ring).

The dimension parity controls the Tate weight of the SD slice: on S^3, $\mathrm{Vol}(S^3) = 2\pi^2$, and the SD coefficients sit in $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ (even-weight pure-Tate sub-ring); on S^5, $\mathrm{Vol}(S^5) = \pi^3$, and the SD coefficients sit in the single slice $\pi^3 \cdot \mathbb{Q}$ (odd-weight pure-Tate slice). Both rings sit in $\mathbb{Q}[\pi] \subset \mathcal{MT}(\mathbb{Q})$; both inherit the F–M classification [25] at the static-substrate sub-case $a(t) \equiv 1$. Closed forms derived by Jacobi ϑ_2, ϑ_3 modular transformation in `debug/sprint_a2_s5_sd_coefficients.py`; memo `debug/sprint_a2_s5_mixed_tate_memo.md` (June 2026) documents the full derivation, numerical verification, and dimension-parity sharpening.

The M1 sub-mechanism transfers to S^5 via the $\mathbb{CP}^2$ base measure on the Hopf-like projection $S^5 \to \mathbb{CP}^2$, $\mathrm{Vol}(\mathbb{CP}^2)/4 = \pi^2/2 \in \mathbb{Q}[\pi]$. Whether the M3 sub-mechanism (vertex parity and Hurwitz at the half-integer shift 5/2) admits a parallel cyclotomic-mixed-Tate-at-level-≤ 4 classification on S^5 is a sub-sprint of identical structural shape to the S^3 argument in §5; we record it as an open sub-question below (Q3').

7.4 M3 on S^5: structural closure, empirical PSLQ deferred

The structural classification "does M3 on S^5 admit a parallel cyclotomic-mixed-Tate-at-level-≤ 4 classification?" is closed POSITIVE by §5.5 above (Sprint A6, 2026-06-03; memo `debug/sprint_a6_m3_s5_mem`
Corollary 5.5 states that $M_3^{(S^5)} \subset \mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4, depth bounded by loop order. The remaining empirical target is specific motivic identification of $S_{\min}^{(S^5)}$ via the decomposition-first pipeline of the S^3 round-2 dossier (weight-grading, t-value expansion, stuffle/parity reduction, then weight-homogeneous level-2 PSLQ on survivors), with the natural prediction $S_{\min}^{(S^5)} \in \mathcal{MT}(\mathbb{Z}[1/2])$ as an identifiable classical element.

7.5 Inner-factor Yukawa content: closed by factorisation + Yukawa-PSLQ

The almost-commutative extension $\mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F$ studied in Paper 32 § VIII.C [13] (Sprint H1 closure) admits a Higgs structure but does not autonomously select the Yukawa coupling matrix. Scoping (Sprint A4, 2026-06-03; memo `debug/sprint_a4_scoping_memo.md`) settles the period-theoretic placement using three theorem-grade prior results:

1. Paper 18 § IV.6 Theorem `thm:eta_trivialization`: on the AC extension, the inner M3 (odd-k) Mellin slot vanishes identically. Inner Mellin reaches only $k \in \{0, 2\}$.
2. Paper 18 § IV.6 Theorem `thm:ac_factorization`: the combined heat trace factorises cleanly into an outer (M2 on S^3) factor and an inner factor, with no mixing.
3. Sprint Yukawa-PSLQ (2026-06-03, `debug/sprint_yukawa_pslq_memo.md`): a 162-cell empirical PSLQ sweep at both $\overline{\mathrm{MS}}\, M_Z$ and unification scales returned zero hits for Yukawa values in low-coefficient pure-Tate periods.

The combined statement: the AC-extension Seeley–DeWitt coefficients factorise as

$$a_k^{\mathrm{AC}} \;=\; \underbrace{a_k^{\mathrm{outer}}}_{\in \bigoplus_j \pi^{2j} \cdot \mathbb{Q}} \;\times\; \underbrace{a^{\mathrm{inner}}(y_1, \ldots, y_g)}_{\in \mathbb{Q}[y_i^{-2s}]}, \tag{39}$$

with outer pure-Tate (M2 on S^3) times inner Dirichlet ring in the Yukawa coupling variables. The inner ring is mixed-Tate over $\mathbb{Q}$ iff $y_i^2 \in \mathbb{Q}$; Sprint Yukawa-PSLQ empirically rejects this at coefficient ceiling $M \leq 10^3$. The inner-factor content therefore lives in Paper 18 § IV.6's sixth tier (parameter-tied Dirichlet, Paper 18 "inner-factor input data"), *categorically disjoint* from M1/M2/M3.

No new sub-mechanism is required. The "genuinely new sub-mechanism" reading of the original open question is closed in the negative: the inner-factor escape route is the already-classified sixth tier, which is structurally compatible with — but not derived from — the master Mellin engine M1/M2/M3.

7.6 Open Q5': a possible deeper M2/M3 cyclotomic unification

Both M2 and M3 sub-mechanisms engage $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$, but via two structurally independent mechanisms (Witt-splitting field for M2 per Remark 4.2; cyclotomic conductor for M3 per Proposition 5.2). Both occurrences trace to the dimension-3 parity of S^3: M2 via the Euclidean signature $(+,+,+,+)$ on $\mathbb{R}^4 \supset S^3$, M3 via the half-integer Camporesi–Higuchi shift 3/2 on $\mathrm{spin}(3) = \mathrm{SU}(2)$. Sprint A7 (2026-06-03; memo `debug/sprint_a7_m2_m3_cyclotomic_memo.md`) returns HALF-STRUCTURAL: the shared field is robust at the number-field level, but no shared Tannakian symmetry is known in the published periods program (Brown [22], Deligne [23], Glanois [24], Fathizadeh–Marcolli [25]).

A nine-sprint scoping arc (2026-06-03/04; six Round-1/2 memos `sprint_q5p_dim_sweep_memo.md`, `sprint_q5p_qsm_litread_memo.md`, `sprint_q5p_tannakian_obstruction_memo.md`, `sprint_q5p_deflatio`
`sprint_q5p_k_slot_tannakian_memo.md`, `sprint_q5p_greenfield_marcolli_transport_memo.md`, plus Round-3 closures `sprint_q5p_r3_hp_star_check_memo.md` and `sprint_q5p_r3_pro_dg_category_memo`
sharpens Q5' along three axes:

Period-ring level (DEFLATED). The M2 $\oplus$ M3 output rings sit inside an explicitly identifiable Tannakian sub-category $M^{\mathrm{GV}} \subset \mathrm{MT}(\mathbb{Z}[i,1/2],4)$ with the single $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q}) \subset G^{M^{\mathrm{GV}}}$ acting compatibly (trivially on the M2 pure-Tate sub-category, non-trivially on the M3 cyclotomic part via χ_{-4}). At this level the ambient motivic Galois group already suffices; no GeoVac-distinguishing Tannakian symmetry is needed on the period output values.

Dg-category level (HP_* / Marcolli–Tabuada route STRUCTURALLY DEAD). The master Mellin engine slot index $k \in \{0,1,2\}$ is TANNAKIAN-INVISIBLE on the ambient M^{GV} (the same period value arises from multiple k-slots). The natural sharpening would be a candidate enriched fiber functor $\omega^{\mathrm{tri}} : \mathrm{dg}(\mathcal{T}) \to \mathrm{Vec}_{\mathbb{Q}} \otimes \mathrm{IndexCat}(\{0,1,2\})$ recording which Mellin slot produced each period, via the Marcolli–Tabuada noncommutative-motives machinery (arXiv:1110.2438 Thm 1.1, arXiv:1112.5422) applied to $\varprojlim_{n_{\max}} \mathrm{dg}(\mathcal{T}_{n_{\max}})$ with Berezin reconstruction maps (Paper 38 §L4). Round-3 closes this route at the source on two independent grounds: (a) $\mathrm{HP}_*(\mathcal{A}_{\mathrm{GV}}^{(n_{\max})})$ is Morita-trivial at every finite cutoff and the pro-limit collapses to $\mathrm{HP}_*(C^\infty(S^3))$, still pure-Tate, with no k-distinguishing content (R3.1); (b) the Berezin maps are completely positive but *not* multiplicative (the approximate-identity error $\gamma_{n_{\max}}$ IS the multiplicativity defect), so they live in a propinquity tunnel-pair category, not a pro-dg category — propinquity convergence and cyclic-homology convergence are categorically orthogonal (R3.2).

Cohomological side (HP^* / cosmic-Galois U^* route VIABLE). R3.1's substantive find: the Chern/Connes character of $(\mathcal{A}_{\mathrm{GV}}, H_{\mathrm{GV}}, D_{\mathrm{GV}})$ is non-trivial and lives in HP^* (cohomology, dual to HP_*); the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ *is* the k-decomposition of this character. The k-slot therefore lives on the cohomological-dual side — wrong side for Marcolli–Tabuada, right side for the Connes–Marcolli cosmic Galois group U^* (arXiv:math/0409306; Connes–Marcolli book 2008 Ch. 4). The U^*-shape acts on counterterm-labelled data in the Connes–Kreimer Hopf algebra of Feynman diagrams; the GeoVac analog acts on Mellin-moment-labelled-by-k data on a Hopf algebra structure that must be constructed. This is the precise mathematical target of the long-range open program.

Bit-exact finite-cutoff witness of the k-slot factorisation (Sprint Q5'-CH-1, June 2026; memo `debug/sprint_q5p_ch1_memo.md`, data `debug/data/sprint_q5p_ch1_data.json`). The master Mellin source $\mathrm{Tr}(D^k\, e^{-tD^2})$ on the truncated Camporesi–Higuchi spectral triple at $n_{\max} \in \{2,3,4\}$ exhibits a bit-exact chirality-parity selection rule: plain trace nonzero for $k \in \{0,2\}$ and identically zero for $k = 1$; supertrace identically zero for $k \in \{0,2\}$ and nonzero for $k = 1$. Every "identically zero" is exact rational 0, computed in `sympy.Rational` throughout the panel and verified for both the diagonal CH Λ and the full graph Dirac $D = \Lambda + \kappa A$ at $\kappa = -1/16$. Two structural facts force the rule: the spectral symmetry $\{\pm(n+1/2)\}$ of the CH spectrum, and the chirality balance $\#\{\chi = +1\} = \#\{\chi = -1\}$ at every $n_{\max}$. Both survive the κA perturbation because the E_1 dipole adjacency A flips parity by ± 1 and respects the chirality-balance structure. This factors the $\mathbb{Z}/3$ master Mellin slot index $k \in \{0, 1, 2\}$ into a $\mathbb{Z}/2$ chirality component (M3 separated from M1+M2) times a heat-kernel-order component within HP_0 (M1 vs M2). The $\mathbb{Z}/2$ chirality component is exactly the standard HP_* grading, hence Tannakian-visible via Marcolli–Tabuada [26]. The heat-kernel-order component within HP_0 — distinguishing M1 ($k = 0$, leading coefficient $\dim \mathcal{H}$) from M2 ($k = 2$, leading coefficient $\mathrm{Tr}(\Lambda^2)$) at the level of the small-t expansion of $\mathrm{Tr}(D^k\, e^{-tD^2})$ — is the genuinely Tannakian-invisible component on the standard fiber functor. Q5' therefore sharpens from "is the $\mathbb{Z}/3$ k-slot Tannakian-relevant?" to: does the candidate enrichment ω^{tri} make the heat-kernel-order grading Tannakian-visible via a $\mathbb{Z}/2 \times \mathbb{Z}$-graded refinement of HP_*? The finite-cutoff symbol level of ω^{tri} is now explicit ($G_0(t)$ lands in M1 with leading $\dim \mathcal{H}$, $G_2(t)$ in M2 with leading $\mathrm{Tr}(\Lambda^2)$, $S_1(t)$ in M3 with leading $\mathrm{Tr}(\gamma\Lambda)$). Lifting this to a graded fiber functor on the dg-category is the natural Stage 1 of the substantial open U^* program; the resulting motivic Galois action is Stage 2.

Continuum M2 verification at integer s (Sprint Q5'-CH-2, 2026-06-05; memo `debug/sprint_q5p_ch2_memo` The continuum CH spectral zeta $\zeta_{D^2}(s)$ for $s \in \{1, 2, 3, 4, 5\}$ sits bit-exactly in $\bigoplus_k \pi^{2k}\mathbb{Q}$ as exactly two non-zero π^{2k} terms per s: $-\pi^2/4$, $\pi^2 - \pi^4/12$, $\pi^4/3 - \pi^6/30$, $\frac{2}{15}\pi^6 - \frac{17}{1260}\pi^8$, $\frac{17}{315}\pi^8 - \frac{31}{5670}\pi^{10}$ respectively. The two-term shape is the integer-s continuum content of the Dirac SD two-term exactness on S^3 (Paper 51 Cor. 2.1) and supplies the continuum bit-exact target of the M2 component of ω^{tri}: for each s, the M2 period content has a $\mathbb{Q}$-fingerprint of two rationals with small denominators, generated by the Hurwitz reduction $\zeta_{D^2}(s) = 2\,\zeta(2s - 2, 3/2) - (1/2)\,\zeta(2s, 3/2)$ via the standard identity $\zeta(s, 1/2) = (2^s - 1)\,\zeta(s)$.

Continuum M3 verification at integer s, with parity-of-s stratification (Sprint Q5'-CH-3, 2026-06-05; memo `debug/sprint_q5p_ch3_memo.md`). The Paper 28 Theorem 3 closed form $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ is verified bit-exact at $s \in \{4, 5, 6\}$ to residual $< 10^{-80}$ at 80 dps (numeric cross-check via Hurwitz $\beta(s) = 4^{-s}(\zeta(s, 1/4) - \zeta(s, 3/4))$). The M3 cyclotomic content shows up bit-exactly only at EVEN s: $s = 2$ gives $2G - 1$, $s = 4$ gives $8\beta_4 - 8G$, $s = 6$ gives $32(\beta_6 - \beta_4)$ — each genuinely in $\mathrm{MT}(\mathbb{Z}[i, 1/2], 4)$ via $\beta(2k)$ for $k \geq 1$ having no known closed form in π. At odd s the M3 expression COLLAPSES to M1 via the Euler-style $\beta(2k+1) = (-1)^k E_{2k}/(2(2k)!) \cdot (\pi/2)^{2k+1}$ closed form: $s = 3$ gives $\pi^3/8 - \pi$, $s = 5$ gives $-\pi^3/2 + 5\pi^5/96$. This sharpens the M2/M3 contrast: M2 is depth-0 uniformly across integer s, M3 is depth-1 stratified by parity of s. Together with Q5'-CH-1 and Q5'-CH-2, the symbol level of the candidate fiber functor ω^{tri} is now explicit on each of M1, M2, M3 with bit-exact targets: M1 via chirality-blind heat trace, M2 via the two-term $\pi^{2k}\mathbb{Q}$ fingerprint, M3 via the parity-stratified cyclotomic content. Stage 1 of the cosmic-Galois U^* program has its complete symbol-level target.

Stage 1 reframe from HP_* *to* HP^* *(Sprint Q5'-Stage1-reframe, 2026-06-05; memo `debug/sprint_q5p_stag`* The R3 sub-sprint of June 2026 closed the Marcolli–Tabuada HP_*-route at source: at every finite $n_{\max}$, the algebra $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} \cong \mathbb{C}^{N_{\mathrm{Fock}}}$ is commutative finite-dimensional, hence Morita-trivial, hence $\mathrm{HP}_*(\mathcal{A}) = (\mathbb{Q}^{N_{\mathrm{Fock}}}, 0)$ with no $\mathbb{Z}/3$ content available for the candidate ω^{tri} enrichment. The reframe identifies the live target as the *cohomological-dual* side HP^*, with the explicit representative being the Jaffe–Lesniewski–Osterwalder (1988) entire-cyclic cocycle $\{\phi_n\}_{n\geq 0}$. The CH-1 bit-exact symbol data — $G_0(t)$ landing in M1 with leading $\dim \mathcal{H}$, $G_2(t)$ in M2 with leading $\mathrm{Tr}(\Lambda^2)$, $S_1(t)$ in M3 with leading $\mathrm{Tr}(\gamma\Lambda)$ — becomes the short-time asymptotic of ϕ_n evaluated on the algebra unit. The S5 falsifier sprint (`debug/sprint_q5p_s5_falsifier_memo.md`, 2026-06-05) independently confirms that the Camporesi–Higuchi substrate is the correct target: the bit-exact $\mathbb{Z}/2$ chirality-parity factorization that grounds CH-1 fails structurally on Bargmann–Segal Hardy-on-S^5 (no $\pm$-symmetric spectrum, no canonical chirality grading on the $(N, 0)$ tower). Stage 2 (the motivic Galois group of the slot-graded Chern-character Tannakian category, shape-precedented by Connes–Marcolli cosmic Galois U^*) remains a long-range open program, unchanged in scope.

Stage 1 explicit ω^{tri} *symbol at* $n_{\max} = 2$ *and cutoff sweep (Sprints Q5'-Stage1-JLO-nmax2 and Q5'-Stage1-2a, 2026-06-05; memos `debug/sprint_q5p_jlo_nmax2_memo.md` and `debug/sprint_q5p_2a_j`*

The JLO entire-cyclic cochains $\{\phi_n^{\rm even}, \phi_n^{\rm odd}\}_{n\in\{0,1,2\}}$ of the truncated CH spectral triple at $n_{\max} = 2$ ($\dim \mathcal{H} = 16$, algebra $\mathbb{C}^5$, Dirac $D = \Lambda + \kappa A$ with $\kappa = -1/16$) have been computed bit-exactly in `sympy.Rational`. The CH-1 leading-coefficient predictions are recovered with a structural shift: M_1 and M_2 live as the t^0 and $-t^1$ coefficients of $\phi_0^{\rm odd}(1;t) = \operatorname{Tr} e^{-tD^2} = 16 - 84\,t + (489/2)\,t^2 - \cdots$; M_3 lives *outside* the JLO framework on commutative $\mathcal{A}$ and inside the Connes–Moscovici residue framework as the leading coefficient of $\operatorname{Tr}(\gamma\, D\, e^{-tD^2}) = 36 - 201\,t + \cdots$. The first non-trivial JLO cochain degree on commutative $\mathcal{A}$ is $n = 2$; structural vanishing $\phi_1(a_0, a_1; t) \equiv 0$ is proved bit-exactly for all idempotent pairs. The bit-exact symbol $\omega^{\rm tri}(\mathcal{T}_2) = (16, 84, 36) \in \mathbb{Z}^3$ extends to $n_{\max} \in \{1, 2, 3, 4\}$ as the polynomial triple

$$
\begin{aligned}
M_1(n_{\max}) &= \tfrac{2\,n_{\max}(n_{\max}+1)(n_{\max}+2)}{3},\\
M_2(n_{\max}) &= \tfrac{n_{\max}(n_{\max}+1)(n_{\max}+2)(2n_{\max}+1)(2n_{\max}+3)}{10},\\
M_3(n_{\max}) &= \tfrac{n_{\max}(n_{\max}+1)^2(n_{\max}+2)}{2},
\end{aligned}
$$

giving panel values $\{(4, 9, 6), (16, 84, 36), (40, 378, 120), (80, 1188, 300)\}$. Each closed form is re-derived from the chirality-balanced shell degeneracy $2n(n+1)$ and the CH eigenvalue $n+1/2$, not curve-fit; no free parameters. The $\omega^{\rm tri}$ symbol at finite cutoff is integer-valued across the pro-system; all transcendentals are deferred to the continuum limit. The mixed JLO + CM-residue framework is the working cochain-level home of $\omega^{\rm tri}$; the single-cocycle vs. multi-cocycle distinction between JLO and CM is itself Stage-2-relevant.

Stage 1 Mellin localization (Sprint Q5'-Stage1-2b, 2026-06-05; memo ***`debug/sprint_q5p_2b_phi0_mellin_`*** The formal Mellin transform of the degree-0 JLO cochain $\mathcal{M}[\phi_0^{\rm odd}(1;t)](s) = \Gamma(s)\cdot\zeta_{D^2}(s)$ closes bit-exactly in $\mathbb{Q}[\Gamma(s)]$ at $n_{\max} \in \{2, 3\}$ for both Λ-only and full $D = \Lambda + \kappa A$. The localization of the master-Mellin partition within this transform sharpens the three sub-mechanisms into three categorically different extraction points: M_1 at the spectral-dimension pole $s = d/2 = 3/2$ as a continuum residue (leading coefficient $\sqrt{\pi}/2$ supplies the Hopf-base measure π); M_2 at integer-s regular points of the same transform (continuum closed forms reproduce the Q5'-CH-2 panel verbatim); M_3 in the η-style pairing $\Gamma(s)\cdot\mathcal{M}[\operatorname{Tr}(\gamma\, D\, e^{-tD^2})](s)$, a structurally different Mellin complex. At finite $n_{\max}$ the spectral zeta is rational-valued (e.g., for the full Dirac at $n_{\max} = 2$, $\zeta^{(2)}_{D^2}(1) = 1\,978\,465\,196\,515\,232/532\,976\,619\,280\,625$); the $s \to 0$ residue is exactly $\dim\mathcal{H} \in \mathbb{Z}$. Master-Mellin transcendentals enter *only* in the $n_{\max} \to \infty$ continuum limit, via the Weyl-law short-time leading $t^{-d/2}$ asymptotic. The cosmic-Galois Stage 2 motivic-Galois action acts at continuum, not at finite cutoff; this is the explicit boundary.

Stage 1 bicomplex and $\mathrm{HP}^{\rm even}$ *class identification (Sprint Q5'-Stage1-2c, 2026-06-05; memo* ***`debug/sprint_q5p_2c_bicomplex_memo.md`****).* The cyclic (b, B) bicomplex of the truncated Camporesi–Higuchi spectral triple at $n_{\max} = 2$ has been constructed bit-exactly in `sympy.Rational`. The load-bearing degree-1 entire-cyclic cocycle condition $b\phi_0 + B\phi_2 = 0$ holds bit-exactly on the full panel of 36 idempotent-pair inputs at three t-orders t^0, t^1, t^2, both even and odd flavors (216 bit-exact zero residuals). The $\mathrm{HP}^{\rm even}$ class on the Morita-trivial baseline $\mathbb{Q}^5 = \mathrm{HP}_0(\mathbb{C}^5)$ is explicitly

$$
[\phi_{\rm JLO}]^{\mathrm{HP}^{\rm even}} \;=\; (+2, -2, +2, +2, -4) \in \mathbb{Q}^5,
$$

the sector-resolved McKean–Singer index summing to $\operatorname{Tr}(\gamma) = 0 = \chi(S^3)$. The Stage-1 symbol $(16, 84, 36)$ is therefore a JLO cocycle *representative*, not a coboundary. Honest scope: at degree 3, the closure $(b\phi_2 + B\phi_4) = 0$ has a bit-exact residual $\pm 1/(3\cdot 2^{16}) = \pm 1/196\,608$ on specific palindromic 4-tuples, a JLO-tower truncation artifact tied to the structural vanishing of odd-degree cochains $\phi_1 \equiv \phi_3 \equiv 0$ on commutative $\mathcal{A}$. The CM-residue cocycle for M_3 is single-cochain and immune to this artifact; the JLO/CM-residue distinction is therefore Stage-2-relevant at the level of *cocycle classes*, not only mechanisms. Stage 1 of the cosmic-Galois U^* bridge is now bit-exactly constructed across four diagnostic axes (symbol, mixed cochain framework, Mellin extraction localization, bicomplex cocycle class). Stage 2 — defining the slot-graded Chern-character Tannakian category and identifying its motivic Galois group — remains a long-range open program, now operating on *two* cocycle classes (JLO $\mathrm{HP}^{\mathrm{even}}$ class hosting M_1, M_2; CM-residue η-class hosting M_3) rather than three slots of a single functor.

Stage 1 second span: explicit CM-residue η-pairing bicomplex and CM η-class identification (Sprint Q5'-Stage1-CM-Bicomplex, 2026-06-05; memo `debug/sprint_q5p_cm_bicomplex_memo.md`). The Connes–Moscovici η-pairing residue cocycle $\psi_n^\eta(a_0,\ldots,a_n;t) = \int_{\Delta_n} \mathrm{Tr}(\gamma D\, a_0\, e^{-s_0 tD^2}\,[D,a_1]\cdots[D,a_n]\, e^{-s_n t}$ on the truncated CH spectral triple at $n_{\max} = 2$ is constructed bit-exactly in `sympy.Rational`; it replaces the JLO leftmost insertion γ with γD and shares the bicomplex structure $(b+B)\psi = 0$ with JLO. The load-bearing degree-1 cocycle condition $b\psi_0^\eta + B\psi_2^\eta = 0$ holds bit-exactly on the full panel of 36 idempotent-pair inputs at three t-orders both flavors (216 bit-exact zero residuals), with *no* truncation artifact of the JLO $\pm 1/196\,608$ type at the load-bearing degree. The CM η-class on the Morita-trivial baseline is explicitly

$$[\psi_{\mathrm{CM}}^{\eta,\mathrm{even}}]^{\mathrm{HP}_\eta} \;=\; (3,3,5,15,10) \in \mathbb{Z}^5,$$

summing to $M_3(n_{\max} = 2) = 36$ (the Sub-Sprint 1 ground truth). The sector-resolved structure is the chirality-symmetrized eigenvalue magnitude per sector $\eta_s = \dim_s \cdot (n_s + 1/2)$, in clean dual to the JLO $\mathrm{HP}^{\mathrm{even}}$ chirality balance $\chi_s = \dim(\gamma_+ e_s \mathcal{H}) - \dim(\gamma_- e_s \mathcal{H})$: both classes decompose over the same K_0 generators $\{[e_s]\}_{s=0..4}$ but encode distinct cohomological content (JLO: zeroth heat-kernel supertrace coefficient = McKean–Singer index $\chi(S^3) = 0$; CM-η: leading t^0 η-density supertrace coefficient $= M_3$). Cross-cutoff sanity at $n_{\max} = 3$: $M_3(3) = 120$ bit-exact, CM-η degree-1 cocycle bit-exact zero on all 9 diagonal sectors. The umbrella memo's structural sketch "M3 lives outside JLO and inside the CM residue framework" is hereby upgraded to a theorem-grade closure at the bicomplex-cocycle-class level.

Stage 1 pro-system functoriality across cutoffs (Sprint Q5'-Stage1-Prosystem, June 2026; memo `debug/sprint_q5p_prosystem_memo.md`, data `debug/data/sprint_q5p_prosystem.json`). The Stage-1 cocycle classes form a strictly compatible inverse system across $n_{\max} \in \{1,2,3,4\}$. Sector idempotents at cutoff $n_{\max}$ are indexed by (n,l) with $1 \le n \le n_{\max}$ and $0 \le l \le n$, giving $N(n_{\max}) = n_{\max}(n_{\max}+3)/2$ sectors $(2,5,9,14)$ nested by inclusion. The truncation $P_{n+1\to n} : \mathcal{O}_{n+1} \to \mathcal{O}_n$ is sector projection (algebra homomorphism); the algebraic Berezin $B_{n\to n+1}$ is sector inclusion. Per-sector closed forms

$$\chi_{(n,l)} = \begin{cases} +2 & l < n \\ -2n & l = n \end{cases}, \qquad \eta_{(n,l)} = \begin{cases} (2l+1)(2n+1) & l < n \\ n(2n+1) & l = n \end{cases},$$

each $n_{\max}$-independent, give pro-system functoriality at the class level bit-exactly: $P^*_{n+1\to n}[\psi^{(n+1)}] = [\psi^{(n)}]$ holds across all three consecutive cutoff pairs for both cocycle classes (six bit-exact zero residuals plus six Berezin compatibility identities). Total sum invariants match the Sub-Sprint 2a polynomial closed forms M_1, M_3 bit-exactly at all four cutoffs; the McKean–Singer index $\sum_s \chi_s = 0$ is preserved cutoff-uniformly. The clean cleanliness of the rule is the cohomological-side trace of the Camporesi–Higuchi shell decomposition's sector-locality: adding a higher shell does not modify the Dirac structure of lower shells, so per-sector cocycle-class values are cutoff-independent functions of (n,l) alone. Closing the strict-strong-form question (pro-system functoriality at the level of *cochain morphisms* in the entire-cyclic bicomplex, rather than at the class level closed here) is sharpened to a bit-exact dichotomy by the strict-strong-form drift diagnostic below; the deep NCG-mathematics gap addressed by Connes 1994 Ch. IV and Loday § 1.4 remains open, orthogonal to and independent of the Stage-1 class-level closure documented here.

Stage 2 first scoping step: candidate Hopf algebra $\mathcal{H}_{\mathrm{GV}}$ (Sprint Q5'-Stage2-Hopf, June 2026; memo `debug/sprint_q5p_stage2_hopf_memo.md`; data `debug/data/sprint_q5p_stage2_hopf.json`). Building on the pro-system substrate above, the first scoping step of Stage 2 produces a concrete bit-exact finite-cutoff candidate Hopf algebra

$$\mathcal{H}_{\mathrm{GV}}^{(n_{\max})} = \mathrm{Sym}_{\mathbb{Q}}\left(\bigoplus_{(n,l)\in\mathcal{S}_{n_{\max}}} \bigoplus_{k\in\{0,1,2\}} \mathbb{Q}\cdot x_{(n,l),k}\right),$$

the polynomial algebra on $3N(n_{\max})$ primitive generators (15 at $n_{\max}=2$, 27 at $n_{\max}=3$), where $k\in\{0,1,2\}$ indexes the Mellin slot. With primitive coproduct, the standard counit $\varepsilon(x_g)=0$, and antipode $S(x_g)=-x_g$, all five Hopf axioms hold bit-exactly at $n_{\max}\in\{2,3\}$ (269 zero residuals on the axiom panel: 52 coassociativity, 104 counit, 104 antipode, 9 bialgebra-compatibility); the Mellin-slot k-grading is preserved by Δ (42 zero residuals), giving the structural product factorisation $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}_{\mathrm{GV}}^{[0]} \otimes \mathcal{H}_{\mathrm{GV}}^{[1]} \otimes \mathcal{H}_{\mathrm{GV}}^{[2]}$; and the pro-system truncation $P_{n+1\to n}$ lifts bit-exactly to a Hopf-algebra homomorphism (126 zero residuals on the truncation panel). The candidate motivic Galois group at finite cutoff is the additive affine group $U^{*(n_{\max})}_{\mathrm{GeoVac}} = \mathbb{G}_a^{3N(n_{\max})}$ with the M1/M3/M2-respecting product structure $\prod_{k=0}^{2}\mathbb{G}_a^{N(n_{\max})}$. Structurally, $\mathcal{H}_{\mathrm{GV}}$ is the *abelian primitive-cospan* of the Connes–Kreimer Hopf algebra of Feynman graphs (Connes–Kreimer 1998, arXiv:hep-th/9808042) — same generator-and-grading shape, same pro-system truncation, but with the sub-graph coproduct of Connes–Kreimer replaced by the primitive coproduct here because the CH Fock sector idempotents are pairwise orthogonal (sector-disjoint, no nesting). The natural long-range continuation for Stage 2's full Tannakian construction (Connes–Marcolli 2007, arXiv:math/0409306; Connes–Marcolli book 2008 Ch. 4) is to enrich the coproduct with a sub-sector or nested-Hopf-tower structure that introduces non-abelian pro-unipotent content. Three candidate structural ingredients are flagged in the memo: nested-Hopf-tower $J^*(S^3)$, cross-shell off-diagonal Dirac perturbation, and JLO/CM bicomplex Cuntz extension. The substrate established here is the concrete object that Stage 2's long-range program acts on.

Stage 1 sub-sprint strict-strong-form pro-system functoriality diagnostic (Sprint Q5'-Stage1-StrictStrong, June 2026; memo `debug/sprint_q5p_strict_strong_memo.md`*; data* `debug/data/sprint_q5p_s` The v3.60.0 pro-system result that the JLO $\mathrm{HP}^{\mathrm{even}}$ class and CM-η class form a strictly compatible inverse system at the level of cocycle CLASSES extends to a sharp dichotomy when probed at the cochain-morphism level. The degree-3 closure residual $(b\phi_2 + B\phi_4)$ at t^0 on the (e_2, e_3) palindromic 4-tuples takes bit-exact rational values $\pm 1/(3\cdot 2^{16})$ at $n_{\max}=2$, transitioning to $\pm 1/2^{16}$ at $n_{\max}=3$ and remaining bit-exactly fixed there for $n_{\max}=4$. The cochain-level pullback $P^*_{3\to 2}[\phi_4^{(3)}]$ does not equal $\phi_4^{(2)}$ on common inputs; the increment $\Delta(B\phi_4) = -1/(3\cdot 2^{14})$ closes the residual identity bit-exactly: $1/196608 + (-1/49152) = -1/65536$. The CM-$\eta$ cocycle has the structurally analogous pattern at degree 3 (drift $-59/(3\cdot 2^{18}) \to \pm 1/2^{13}$, fixed-point from $n_{\max}=3$ onward), confirming that the Track 1 clean-cochain identification of CM-η is restricted to degree 1. Both cocycle towers therefore fail strict cochain-morphism functoriality on commutative $\mathcal{A}$ at every tested cutoff $n_{\max}\geq 2$. The pro-system verdict at the cochain-morphism level is bit-exact non-strict with $n_{\max}=3$ fixed-point structural drift; the verdict at the class level is bit-exact strict (v3.60.0). This sharp dichotomy constitutes the bit-exact empirical content of the published-open Connes 1994 Ch. IV / Loday §1.4 gap on entire-cyclic functoriality under algebra truncations, made explicit on the truncated CH commutative subalgebra in the GeoVac substrate.

Tauberian rate uniformity at $s=3/2$ closed (Sprint Q5'-Stage1-Tauberian, 2026-06-05; memo `debug/sprint_q5p_tauberian_memo.md`*; data* `debug/data/sprint_q5p_tauberian.json`*).* The Stage-1 named follow-on on uniform rate of approach to the M1 Mellin residue at $s=3/2$ is closed at Paper 38 L2 grade. The four hypotheses of the standard uniform Tauberian theorem (Karamata [27] § V.3; Korevaar [28] § III.4; Tenenbaum [29] § II.7 Thm II.7.7) transport bit-exactly: positivity, polynomial growth, simple-pole meromorphic continuation (v3.59.0 Track 2), vertical-strip boundedness. On the compact above-pole neighborhood $U_> = [3/2+\epsilon, 3/2+1/4]$ the remainder $\sup_{s\in U_>} |\zeta^{(N)}_{D^2}(s) - \zeta^{\mathrm{cont}}_{D^2}(s)|$ decays uniformly as $O(N^{-2\epsilon+\delta})$. Boundary-saturation verified bit-exactly: sup attained at $s = 25/16$ (closest grid point above pole) at every cutoff $n_{\max} \in \{2, 3, 4, 5\}$; empirical decay slope -0.092 at $n_{\max} \in \{2, \ldots, 5\}$ monotonically approaches the predicted $-1/8 = -0.125$, hitting -0.123 at $n_{\max} = 100$.

Stage 1 third span: continuum M3 Mellin residue and integer-s panel (Sprint Q5'-Stage1-M3-Continuum, 2026-06-05; memo `debug/sprint_q5p_m3_continuum_memo.md`*; data* `debug/data/sprint_q5p_m3` Closes the named M3 continuum-residue follow-on of v3.59.0 Track 2 at Paper 38 L2 grade. The continuum η-Dirichlet series $\eta_D(s) = \sum_{n \geq 0} g_n |\lambda_n|^{1-s}$ has the Hurwitz reduction $\eta_D(s) = 2\zeta(s-3, 3/2) - \frac{1}{2}\zeta(s-1, 3/2) = D(s-1)$ (Paper 28's Dirac Dirichlet series), with TWO simple poles at $s = 4$ and $s = 2$ of bit-exact meromorphic residues 2 and $-1/2$ respectively (two independent routes: Hurwitz pole rule + direct Laurent via digamma). Mellin residues $\mathrm{Res}_{s=4}\Gamma(s)\eta_D(s) = 12$ and $\mathrm{Res}_{s=2}\Gamma(s)\eta_D(s) = -1/2$ are rational-integer; the genuine M3 cyclotomic content lives in the chirality-resolved $(D_{\mathrm{even}} - D_{\mathrm{odd}})(s-1)$ via the $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$ Galois descent (Q5'-CH-3 panel reproduced verbatim at $s \in \{3, 5, 7\}$: $-1+2G$, $8\beta_4 - 8G$, $-32\beta_4 + 32\beta_6$). Discrete-side Karamata Tauberian signature on the v3.60.0 substrate: $S(n_{\max})/\log(n_{\max}) \to 2$ at $n_{\max} \geq 200$ matches the residue exactly, with no Jacobian correction (natural η-Mellin variable IS the spectral $|\lambda| = n + 1/2$). *Structural finding (M1/M3 asymmetry IS the operational content of the master-Mellin partition):* unlike M1 (three exact-factor sibling normalizations π, $4/\pi$, $\sqrt{\pi}/2$ of a single generator; Paper 18 §III.7 / Paper 38 §VIII / v3.59.0 Track 2), M3's ring $\mathcal{MT}(\mathbb{Z}[i, 1/2], 4)$ is depth-graded with successive generators $\{1, G, \beta_4, \beta_6, \ldots\}$ and no π-power reduction at any depth ≥ 1 (Apéry-style irrationality). M1 lives at $k = 0$ depth 0 pure-Tate; M3 lives at $k = 1$ depth 1+ cyclotomic level 4.

Substrate enrichment: nested-Hopf-tower $J^*(S^3)$ (Sprint Q5'-J-Star-S3, June 2026; memo `debug/sprint_q5p_j_star_s3_memo.md`*; data* `debug/data/sprint_q5p_j_star_s3.json`*).* A first scoping step of the first of three long-range enrichment ingredients flagged in the Stage-2 Hopf paragraph above replaces the sector-disjoint $\mathbb{C}^{N(n_{\max})}$ substrate with the SU(2) Peter–Weyl matrix-coefficient substrate $J^*_{j_{\max}}(S^3)$. The matrix-coefficient coproduct $\Delta\pi^j_{mn} = \sum_p \pi^j_{mp} \otimes \pi^j_{pn}$ is bit-exactly non-primitive on every $j > 0$ generator at $j_{\max} \in \{1/2, 1, 3/2\}$; coassociativity and counit pass bit-exactly in the free polynomial algebra (147 zero residuals); antipode axiom holds at the standard $\mathcal{O}(SU(2)) \cong \mathcal{O}(SL_2)$ quotient by SU(2) unitarity relations; pro-system truncation lifts to a Hopf-algebra homomorphism (132 zero residuals). The candidate motivic Galois group at the quotient is SL_2 at every $j_{\max} \geq 1/2$ — non-abelian semisimple algebraic group. The enrichment trade is axis-orthogonal: $J^*(S^3)$ has non-abelian content but no Mellin-slot k-partition; v3.61.0's $\mathcal{H}_{\mathrm{GV}}$ has the Mellin partition but is abelian. Natural long-range synthesis: tensor product giving $U^* = \mathbb{G}_a^{3N} \times SL_2$ (Levi-decomposition shape, the right Stage-2 target).

Substrate enrichment: cross-shell off-diagonal Dirac (Sprint Q5'-OffDiag-Dirac, June 2026; memo `debug/sprint_q5p_offdiag_dirac_memo.md`*; data* `debug/data/sprint_q5p_offdiag_dirac.json`*).* The first scoping step of the second enrichment ingredient (cross-shell off-diagonal Dirac perturbation via E_1-dipole transition generators $T_{s' \to s} := e_s \cdot (\kappa A) \cdot e_{s'}$ in the CH spectral triple at $n_{\max} = 2$) closes POSITIVE. The enriched substrate has 5 idempotent + 12 single-step transition generators with non-trivial matrix-algebra relations: all 12 transitions carry non-vanishing η-class values in $\kappa^2 \cdot \mathbb{Z}$ (sector-locality breakdown), 12 chain compositions land on idempotents forcing non-primitive coproduct cross-terms with bit-exact non-zero $\eta \otimes \eta$ image, and 18 chain compositions land on two-step transitions outside the basic basis (algebra-closure failure — the load-bearing long-range follow-on). 24 of 66 = 36% of transition-pair commutators are non-zero, giving explicit non-abelian Lie generators. *Structural alignment with v3.61.0 Track B drift:* the non-primitivity content has denominator family $\kappa^2 \cdot \mathbb{Z} = 2^{-8} \cdot \mathbb{Z}$ times inverse path counts, structurally aligned with the bit-exact $\pm 1/2^{16}$ closure drift residual of v3.61.0 Track B on (e_2, e_3) palindromes modulo a 2^3 JLO simplex normalization factor — supporting at the structural level the conjecture from the v3.61.0 umbrella that the two findings of v3.61.0 are aspects of one Stage-2 substrate-enrichment story. Long-range next steps: algebra-closure dimension at general $n_{\max}$, closed-form Lie structure constants, bit-exact bridge identification.

Substrate enrichment: Cuntz extension is structurally incompatible (Sprint Q5'-Cuntz, June 2026; memo `debug/sprint_q5p_cuntz_memo.md`*; data* `debug/data/sprint_q5p_cuntz.json`*).* Of the three substrate-enrichment ingredients flagged in the Stage-2 Hopf paragraph above, the first scoping step of the third returns a structural negative. Replacing the commutative algebra $\mathbb{C}^{N(n_{\max})}$ of sector idempotents by the Cuntz algebra $\mathcal{O}_N$ generated by $N = 3N(n_{\max})$ isometries $S_{(n,l),k}$ produces a non-commutative substrate, but both natural coproduct candidates fail compatibility with the Cuntz defining relations bit-exactly: the primitive coproduct $\Delta(S_i) = S_i \otimes 1 + 1 \otimes S_i$ fails $S_i^* S_j = \delta_{ij}$ on 0/225 pairs at $n_{\max} = 2$, and the diagonal coproduct $\Delta(S_i) = S_i \otimes S_i$ passes the first Cuntz relation but fails $\sum_i S_i S_i^* = 1$ by exactly $N(N-1) = 210$ missing cross-terms. Structurally, $\mathcal{O}_N$ is a Hopf-*comodule* over gauge $U(1)$ rather than a Hopf-algebra (Pimsner–Voiculescu 1980, *J. Operator Theory* 4; Pimsner 1997, Fields Inst. Commun. 12; Connes–Cuntz 1988, Commun. Math. Phys. 114). The M1/M2/M3 partition survives only at the trivial comodule level via fixed-k generator labels; the clean Hopf tensor product established for the commutative case is structurally weakened. *The Cuntz route is the structurally most disruptive of the three enrichment ingredients; the remaining two preserve the Hopf-algebra category and are correspondingly more tractable.* Long-range continuation via Pimsner–Voiculescu sequence.

Stage 2 HEADLINE substrate: Levi-decomposition tensor synthesis (Sprint Q5'-Levi-Synthesis, June 2026; memo `debug/sprint_q5p_levi_synthesis_memo.md`*; data* `debug/data/sprint_q5p_levi_synthe`
Combining the v3.61.0 abelian primitive substrate with the v3.62.0 Peter–Weyl substrate via tensor product $\mathcal{H}_{\mathrm{GV}}^{\mathrm{Levi},(n_{\max},j_{\max})} := \mathcal{H}_{\mathrm{v3.61}}^{(n_{\max})} \otimes_{\mathbb{Q}} \mathcal{H}^{J^*,j_{\max}}$ gives the HEADLINE Stage-2 substrate. At finite cutoff $(n_{\max}, j_{\max}) \in \{(2,1/2),(2,1),(3,1/2)\}$, all five Hopf axioms hold bit-exactly with k-grading preserved on the first factor and non-cocommutativity preserved on the second — 882 bit-exact zero residuals (636 axiom + 246 truncation). The candidate motivic Galois group at the standard $\mathcal{O}(SL_2)$ quotient is $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N(n_{\max})} \times SL_2$ via the Tannakian theorem (Waterhouse 1979 §1.4; Sweedler 1969 Ch. IV; Deligne 1990). The semidirect product reduces to a direct product because the Levi action on the unipotent radical is trivial: v3.61.0's generators carry no $SU(2)$ representation content, so any SL_2 element acts as identity (Hochschild 1981 Ch. VII). Dimensions: 18 at $n_{\max} = 2$, 30 at $n_{\max} = 3$ (independent of $j_{\max} \geq 1/2$ because the Peter–Weyl filtration is internal to $\mathcal{O}(SL_2)$). The Levi-decomposition shape (pro-unipotent times semisimple) is the published structural target for the cosmic-Galois U^* in Connes–Marcolli's motivic Galois machinery (arXiv:math/0409306; book 2008 Ch. 4). *The Stage-2 substrate-construction phase of the Q5' program closes here.* Long-range continuation: (a) full Tannakian closure of the pro-unipotent factor; (b) verification that the cosmic-Galois U^* acts on $\mathcal{H}_{\mathrm{Levi}}$ in the expected way.

Mellin-slot-decorated Peter–Weyl substrate: alternative synthesis $U^* = SL_2^3$ (Sprint Q5'-Decorated-PW, June 2026; memo `debug/sprint_q5p_decorated_pw_memo.md`*; data* `debug/data/sprint_q5p_`
The alternative synthesis flagged in the J*(S^3) paragraph above option (b) — the Mellin-slot-decorated Peter–Weyl substrate $\pi^{j,k}_{mn}$ at $j_{\max} \in \{1/2, 1\}$ with $n_k = 3$ Mellin slots and k-preserving coproduct — passes all Hopf axioms bit-exactly with the standard counit and per-slot $SU(2) \to SL_2$ quotient (411 zero residuals). The substrate factorises as $\mathcal{O}(SL_2)^{\otimes 3}$, giving candidate motivic Galois group $U^* = SL_2^3$ with the three master Mellin engine mechanisms M1, M2, M3 realised as three independent non-abelian semisimple SL_2 factors. The alternative k-summing coproduct ($\mathbb{Z}/3$-graded) breaks the antipode axiom at the per-slot $SU(2)$ quotient by a factor of $n_k = 3$, requiring $1/n_k \in \mathbb{Q}(1/3)$ to fix and breaking the discrete-for-skeleton discipline. The two synthesis routes (L1 Levi-decomposition $\mathbb{G}_a^{3N} \times SL_2$ vs L2 decorated SL_2^3) capture different aspects of the Stage-2 substrate: the tensor synthesis treats v3.61.0's abelian primitives as independent content with explicit M-slot grading; the decorated-PW treats the M1/M2/M3 partition as three independent SL_2 symmetries. Both are valid candidates; the strategic choice is the next sprint-scale decision in the long-range Stage-2 program.

Combined substrate scoping: L5 = L1 categorically (Sprint Q5'-Combined-Substrate, June 2026; memo ***debug/sprint_q5p_combined_substrate_memo.md****; data* ***debug/data/sprint_q5p_combined_su*** The first long-range scoping step of combining the T3a Peter–Weyl substrate and the T3b cross-shell off-diagonal Dirac substrate on a SINGLE combined substrate closes the categorical question: **at the basic substrate level, L5 reduces to L1 — the categorical tensor product**. Two clean structural obstructions block the smash-product upgrade: (i) Peter–Weyl matrix coefficients are polynomial *functions* on SL_2 (in $\mathcal{O}(SL_2)$); OffDiag transitions are *operators* on the CH Hilbert space with scalar cocycle data — categorically different objects, no consistent identification at the panel level; (ii) the natural $SU(2)$ z-rotation does NOT preserve the basic 5-sector OffDiag basis bit-exactly (symbolic computation produces three distinct phase factors $\{1, e^{i\theta}, e^{-i\theta}\}$, so the conjugate lives in m_J-resolved sub-transitions outside the basic basis). This *confirms* L1's Levi-decomposition $\mathbb{G}_a^{3N} \times SL_2$ as the right Stage-2 target (no richer smash-product structure available at the basic substrate level), ruling out a category of long-range continuations. Genuinely new content (a non-trivial smash product) lives at the m_J-resolved refinement of the OffDiag basis, sacrificing the v3.61.0 sector structure; long-range.

Bridge identity bit-exactly closing the T3b umbrella (Sprint Q5'-Bridge-Id, June 2026; memo ***debug/sprint_q5p_bridge_id_memo.md****; data* ***debug/data/sprint_q5p_bridge_id.json****).* The bit-exact bridge identification flagged as a feasible follow-on in the Q5'-OffDiag-Dirac paragraph above closes via the single identity $\mathrm{drift}_{n_{\max}\geq 3}(e_2, e_3, e_3, e_2) = -\kappa^4 = -1/2^{16} = -1/65536$, with four bit-exact equivalent forms: direct, via OffDiag $\eta \otimes \eta$ as $-(5/8192)/40$, via JLO simplex times interior path count $-(1/4!) \cdot \kappa^4 \cdot 24$, and as boundary cutoff $+(1/4!) \cdot \kappa^4 \cdot 8 = +\kappa^4/3$. Three named structural ingredients: JLO simplex factor $1/4! = 1/(3 \cdot 2^3)$ (the 2^3 piece reconciling 2^{13} vs 2^{16}); Dirac off-diagonal weight $\kappa^4 = 1/2^{16}$; integer two-step path counts $T^{\mathrm{int}}_{\mathrm{path}} = 24$ (interior), $T^{\mathrm{bdy}}_{\mathrm{path}} = 8$ (boundary). Bit-exact integer-lattice arithmetic: boundary-vs-interior $-3 = 24/8$, pullback closure $32 = 24 + 8$, T3b numerator $5 = 40/8$ where $40 = 2 \cdot 10 \cdot 2$. *Track A v3.61.0 + Track B v3.61.0 + T3b v3.62.0 are all aspects of the same κ^4 structure*, joined explicitly by integer two-step E1 path counts on the OffDiag substrate. Long-range follow-ons (closed-form $T_{\mathrm{path}}(n_{\max})$, Lie-algebra structure constants, continuum-limit Mellin lift to M3) remain open.

OffDiag algebra-closure dimension scaling at $n_{\max} = 3$ (Sprint Q5'-OffDiag-Closure-nmax3, June 2026; memo ***debug/sprint_q5p_offdiag_closure_memo.md****; data* ***debug/data/sprint_q5p_offdiag_c*** The first long-range scoping step of OffDiag algebra-closure dimension at general $n_{\max}$ closes POSITIVE. The closure $\mathcal{A}^{(n_{\max})}_{\mathrm{OD}} = \mathbb{Q}\langle e_s, A_{\mathrm{off}}\rangle$ is bit-exactly the path algebra of the sector quiver realised in $M_{\dim H}(\mathbb{Q})$ via the canonical adjacency representation $A_{\mathrm{off}} = D - \Lambda = \kappa A_{\mathrm{graph}}$: $\dim \mathcal{A}^{(2)}_{\mathrm{OD}} = 84$ and $\dim \mathcal{A}^{(3)}_{\mathrm{OD}} = 592$, decomposing as a sum of per-block ranks over all $N(n_{\max})^2$ sector pairs. Fill-fractions $84/256 = 21/64$ and $592/1600 = 37/100$ slowly increasing (asymptotic limit requires $n_{\max} = 4$); Lie subalgebra of single-step transitions does NOT close (82 of 378 commutator pairs outside the 28-dim transition span at $n_{\max} = 3$). Named sprint-scale follow-on: $n_{\max} = 4$ extension. Long-range: closed-form per-block rank from quiver combinatorics + embedding of the $\mathcal{A}^{(n_{\max})}_{\mathrm{OD}}$ inductive system into the pro-unipotent factor of the cosmic-Galois U^* Hopf algebra (Connes–Marcolli 2007).

Stage-2 substrate decision via the primitive-space invariant: L1 is forced (Sprint Q5'-L1-vs-L2-Diagnostic, June 2026; memo ***debug/sprint_q5p_l1_vs_l2_diagnostic_memo.md****; data* ***debug/data/sprint_q5p_l1_vs_l2_diagnostic.json****).* The strategic question flagged in the decorated-PW paragraph above and in the L1+L5 categorical-reduction paragraph – whether the v3.61.0 abelian primitives $x_{(n,l),k}$ carry independent content (so L1's Levi-decomposition is forced) or are reducible to Peter–Weyl matrix coefficients in disguise (so L2's SL_2^3 substrate suffices) – is resolved bit-exactly at finite cutoff by the primitive-space invariant $\dim \mathrm{Prim}(\mathcal{H}) = \dim \mathrm{Lie}(\mathrm{Spec}\, \mathcal{H})$, a Hopf-isomorphism invariant. Milnor–Moore for the connected cocommutative L1 substrate gives $\dim \mathrm{Prim}(\mathcal{H}_{\mathrm{GV}}) = 3N(n_{\max}) \in \{6, 15, 27, 42, 60, 81\}$ at $n_{\max} \in \{1, \ldots, 6\}$; the standard identification $\mathrm{Prim}\, \mathcal{O}(SL_2)^{\otimes 3} = \mathfrak{sl}_2^3$ gives $\dim \mathrm{Prim}(\mathcal{H}_{\mathrm{dec}}) = 9$ independent of $j_{\max}$ (higher $j_{\max}$ adds degree-≥ 2 symmetric-square Clebsch–Gordan polynomials in the $j = 1/2$

generators, contributing zero new dimensions to $\mathfrak{m}/\mathfrak{m}^2$). At every $n_{\max} \geq 2$ the dim defect is strictly positive $(6, 18, 33, 51, 72)$; the two Hopf algebras cannot be isomorphic. The substrate-dim coincidences $15 = 15$ at $(n_{\max}, j_{\max}) = (2, 1/2)$ and $42 = 42$ at $(4, 1)$ are small-cutoff artifacts breaking at $n_{\max} \in \{1, 3, 5, 6\}$. An independent invariant (cocommutativity) separates the substrates at every $n_{\max} \geq 1$. Bit-exact panel: 258 zero residuals / consistent checks. The L2 SL_2^3 substrate remains a valid complementary semisimple-only reading of the M1/M2/M3 partition; the Levi-decomposition $U^*_{\text{GeoVac,Levi}} = \mathbb{G}_a^{3N(n_{\max})} \times SL_2$ is the structurally-correct Stage-2 target, matching the published Connes–Marcolli motivic-Galois shape. The canonical long-range next step of the Q5' program is the Tannakian closure of the pro-unipotent factor $\mathbb{G}_a^{3N(n_{\max})}$ in the Connes–Marcolli machinery.

Closed-form generating function for the chirality-weighted two-step path count on the OffDiag substrate (Sprint Q5'-T-Path-Generating T2, June 2026; memo `debug/sprint_q5p_t_path_generating_memo` data `debug/data/sprint_q5p_t_path_generating.json`). The chirality-weighted two-step path count $N^{(2)}_{s' \to s} := \mathrm{Tr}(\gamma A e_s A e_{s'})/\kappa^2$ on the v3.62.0 OffDiag substrate is bit-exactly *cutoff-independent* (every realised sector pair has the same value at every $n_{\max}$ where both sectors are present), satisfies the E1 selection rule $|\Delta n|, |\Delta l| \leq 1$, has total signed sum vanishing identically ($T^{\text{signed}}_{\text{path}} \equiv 0 = \mathrm{Tr}(\gamma A^2)/\kappa^2$, the McKean–Singer extension to the chirality-weighted A^2 trace), and obeys the closed form

$$T^{\text{abs}}_{\text{path}}(n_{\max}) \;=\; \frac{(n_{\max}^2 + 9n_{\max} - 4)(n_{\max}^2 + 13n_{\max} - 6)}{6}$$

at panel $(8, 72, 224, 496, 924)$ for $n_{\max} \in \{1, 2, 3, 4, 5\}$. This closed form unlocks the continuum Mellin lift below.

Continuum Mellin lift of the discrete bridge identity to the spectral zeta side with bit-exact M2/M3 decomposition by shift parity (Sprint Q5'-Mellin-Lift-Bridge T1, June 2026; memo `debug/sprint_q5p_mellin_lift_bridge_memo.md`; data `debug/data/sprint_q5p_mellin_lift_bridge.js` Using T2's quartic closed form, the continuum Mellin lift of the v3.63.0 L3 discrete bridge identity $\text{drift}_{n_{\max} \geq 3} = -\kappa^4$ is the generating function

$$F(s) \;=\; \tfrac{1}{6}\zeta(s-4) + \tfrac{11}{3}\zeta(s-3) + \tfrac{107}{6}\zeta(s-2) - \tfrac{53}{3}\zeta(s-1) + 4\,\zeta(s),$$

admitting the bit-exact decomposition by shift parity $F = F_{\text{M2}} + F_{\text{M3}}$ with $F_{\text{M2}}(s) = \frac{1}{6}\zeta(s-4) + \frac{107}{6}\zeta(s-2) + 4\,\zeta(s)$ (even shifts, M2 Seeley–DeWitt) and $F_{\text{M3}}(s) = \frac{11}{3}\zeta(s-3) - \frac{53}{3}\zeta(s-1)$ (odd shifts, M3 vertex-parity-Hurwitz). Five simple poles at $s \in \{1, 2, 3, 4, 5\}$ with residues $\{4, -53/3, 107/6, 11/3, 1/6\}$; integer-$s$ panel exhibits mixed M2 (π^{2k}) + M3 ($\zeta(\text{odd})$) content at every test point (e.g. $F(6) = -53\zeta(5)/3 + \pi^2/36 + 4\pi^6/945 + 11\zeta(3)/3 + 107\pi^4/540$). F is structurally distinct from the v3.62.0 M3 $\eta_D(s)$ Hurwitz Mellin (shares only the odd-shift structure; $\mathbb{Q}$ vs $2^s\,\mathbb{Q}$ coefficients). This is the *first algebraic-side realisation of the M1/M2/M3 master Mellin engine partition appearing on a single Hopf-algebra-level invariant* — the master Mellin engine extends from the trace-evaluation level (case-exhaustion theorem in Paper 32 §VIII) to the substrate-side Mellin generating function.

m_J-resolved smash-product foundation: bit-exact $\Delta m_J \in \{-1, 0, +1\}$ grading and U(1) z-rotation diagonalisation at $n_{\max} = 2$ (Sprint Q5'-mJ-Smash-Product T4, June 2026; memo `debug/sprint_q5p_mj_smash_product_memo.md`; data `debug/data/sprint_q5p_mj_smash_product.json`). The m_J-resolved refinement of the OffDiag substrate that v3.63.0 L5 explicitly named as the load-bearing data for a non-trivial smash-product admits a bit-exact $\mathbb{Z}_3$-grading by $\Delta m_J = m_{J,\text{to}} - m_{J,\text{from}} \in \{-1, 0, +1\}$ (Wigner–Eckart selection for the vector operator $A = \kappa A_{\text{graph}}$) with symmetric dimension count 30/40/30 over 100 non-zero off-diagonal Dirac entries at $n_{\max} = 2$. The U(1) z-rotation $U_\theta|n, \kappa, m_J\rangle = e^{i m_J \theta}|n, \kappa, m_J\rangle$ acts diagonally: $U_\theta T_{m'_J \to m_J} U_\theta^{-1} = e^{i\Delta m_J \theta} T_{m'_J \to m_J}$, reproducing exactly L5's three phase factors $\{1, e^{i\theta}, e^{-i\theta}\}$ — the m_J-resolved refinement *diagonalises* the U(1) action that L5 identified as broken on the basic substrate.

Critical scope finding: the j-content at $n_{\max} = 2$ includes $j = 3/2$ states (from $\kappa = \pm 2$ in sectors $(2,1)$ and $(2,2)$), so the L2 decorated-PW substrate at $j_{\max} = 1/2$ is *insufficient* for the full SU(2) action on the m_J-resolved OffDiag; L2 must be extended to $j_{\max} \geq 3/2$ as the sprint-scale prerequisite to the long-range smash-product construction $\mathcal{A}^{m_J\text{-OD}} \rtimes \mathcal{O}(SL_2)$ — the only path to a non-trivial Hopf extension beyond Levi-decomposition.

L2 substrate extended to $j_{\max} = 3/2$: smash-product prerequisite closed (Sprint Q5'-FO1 v3.66.0; memo `debug/sprint_q5p_fo1_l2_extension_jmax32_memo.md`; data `debug/data/sprint_q5p_fo1_` The sprint-scale prerequisite identified above is closed bit-exactly: the L2 decorated-PW substrate $\mathcal{H}_{\text{dec}}^{(j_{\max}=3/2)}$ has dimension $90 = 3 \cdot 30$ (per-slot dim $\sum_{2j \leq 3}(2j+1)^2 = 1+4+9+16 = 30$), with all 7 Hopf axioms passing: coassociativity (90/90), counit L+R (180/180), antipode at $SU(2)^{\otimes 3}$ quotient (90/90 structural per slot), k-grading preservation (90/90), pro-system Hopf-hom $P_{3/2\to 1}$ (126/126), pro-system $P_{3/2\to 1/2}$ (45/45). Total bit-exact zero residuals: 621. The substrate factorisation $\mathcal{O}(SL_2)^{\otimes 3}$ and candidate motivic Galois group SL_2^3 are unchanged from the v3.63.0 panel at $j_{\max} \in \{1/2, 1\}$ — $j_{\max}$ controls only which irrep matrix coefficients are explicitly included; the underlying group is $j_{\max}$-invariant. This unblocks the long-range smash-product construction at the prerequisite level.

Interpretation C of cosmic-Galois U^*-action closes via period-pairing: $F(s)$ in $MT(\mathbb{Q},1) \subset MT(\mathbb{Z}[i,1/2],4) + \chi, \eta$ at depth 0 (Sprint Q5'-FO2+FO3 v3.66.0; memo `debug/sprint_q5p_fo2_fo3_mt_peri` data `debug/data/sprint_q5p_fo2_fo3_mt_period.json`). T5 of v3.65.0 identified three interpretations of "U^* acts on $\mathcal{H}_{\text{Levi}}$": (A) Hopf-coaction, (B) Hopf-automorphism, (C) period-pairing. Interpretation C is the cleanest sprint-scale closure; this follow-on closes it bit-exactly for all named cocycle-class families. T5 SQ1+SQ2+SQ5 combined: (i) every $\chi_{(n,l)}$ and every $\eta_{(n,l)}$ value (60 + 60 entries up to $n_{\max} = 4$) is an integer in $\mathbb{Z}$ at depth 0; (ii) every term of $F(s)$ at integer $s \in \{6,7,8,9,10\}$ (5 cells × 5 terms = 25 term classifications) is either pure-Tate $\pi^{2k}\mathbb{Q}$ (M2) or odd-Riemann $\zeta(\text{odd})\mathbb{Q}$ (M3); both sit in MT$(\mathbb{Q},1)$. The Paper 18 §III.7 master Mellin engine partition aligns bit-exactly with the MT depth/weight grading at integer-shifted ζ values: M1 (π, Hopf-base measure, depth 0 weight 2), M2 (π^{2k}, Seeley–DeWitt, depth 0 weight $2k$), M3 ($\zeta(\text{odd})$, vertex-parity-Hurwitz, depth 1 weights $3, 5, \ldots$). The cosmic-Galois U^* acts on χ, η trivially (depth 0 fixed-points), on the F_{M2} component as the Tate subgroup (invariant up to π^{2k} scaling), and on the F_{M3} component as the standard motivic Galois action on MT$(\mathbb{Q},1)$ odd-zeta classes (Brown 2012, Glanois 2015). Total bit-exact zero residuals: 145 (25 + 60 + 60). **Interpretation C closes bit-exactly**; Interpretations A (long-range, requires explicit $\mathcal{O}(U^*)$ presentation) and B (sprint-scale via T5 SQ3 Aut_{Hopf} enumeration, not done in this sprint) remain open as named follow-ons.

Pro-system lockdown PS-1: transition maps as closed-form algebra homomorphisms with bit-exact cofiltered axiom (Sprint Q5'-ProSystem-Lockdown, PS-1 sub-track, 2026-06-06; memo `debug/sprint_q5p_ps1_transitions_memo.md`; data `debug/data/sprint_q5p_ps1_transitions.json`; module `geovac/pro_system.py`). The transition maps $P_{m,k} : \mathcal{O}_m \to \mathcal{O}_k$ between truncated sector-idempotent algebras are realised as closed-form $N(k) \times N(m)$ 0/1 integer matrices in canonical lex order on sectors: $P_{m,k} = [I_{N(k)} \mid 0_{N(k)\times(N(m)-N(k))}]$, sending $e_{(n,l)} \mapsto e_{(n,l)}$ if $n \leq k$ and 0 otherwise. The cofiltered axiom $P_{m,k} = P_{n,k} \cdot P_{m,n}$ is verified bit-exact for every triple $1 \leq k < n < m \leq 5$ (10 matrix-level identities, 10 bit-exact zero residuals), promoting the v3.60.0 result (pull-back compatibility at consecutive cutoffs) to a closed-form inverse system in the strict categorical sense. The per-sector closed forms $\chi_{(n,l)}, \eta_{(n,l)}$ extend bit-exact to the new cell $n_{\max} = 5$ (six new sectors, 20 total), with all-pairs pull-back $P^*_{m,k}[\psi^{(m)}] = [\psi^{(k)}]$ holding bit-exact for the 10 pairs $\{(m,k) : 1 \leq k < m \leq 5\}$ and both characters χ, η (20 bit-exact zero residuals at the class-vector level). Total bit-exact zero residuals across the panel: $100 + 10 + 20 = 130$. PS-1 is the substrate layer of the pro-system lockdown program; PS-2 will verify that the cosmic-Galois $U^* = \mathbb{G}_a^{3N(n_{\max})} \rtimes SL_2$ acts compatibly with $P_{m,k}$ for each generator on each of the three Interpretation-C-closed characters $\chi, \eta, F(s)$; PS-3 constructs the inverse limit $\mathcal{O}_\infty = \varprojlim \mathcal{O}_{n_{\max}}$ and extends the U^*-action to the limit; PS-4 probes endomorphism rigidity and Tannakian readiness.

Pro-system lockdown PS-2: U^*-action compatibility with PS-1 transitions (Sprint Q5'-ProSystem-Lockdown, PS-2 sub-track, 2026-06-06; memo `debug/sprint_q5p_ps2_ustar_compatibility_memo.md`; data `debug/data/sprint_q5p_ps2_ustar_compatibility.json`). The block-diagonal Hopf-hom lift $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ of PS-1's $P_{m,k}$ acts on the primitive generators by $\Phi(x^{(m)}_{(n,l),s}) = x^{(k)}_{(n,l),s}$ if $n \le k$ and 0 otherwise, where $s \in \{0,1,2\}$ indexes the master Mellin engine slots (M1, M3, M2 per Paper 18 §III.7). The $3N(k) \times 3N(m)$ matrix of $\Phi_{m,k}$ is the block-diagonal $\mathrm{diag}(P_{m,k}, P_{m,k}, P_{m,k})$ — three independent copies of PS-1's transition, one per Mellin slot. Three bit-exact verifications at $n_{\max} \le 5$: (i) the cofiltered axiom $\Phi_{m,k} = \Phi_{n,k} \cdot \Phi_{m,n}$ holds bit-exact at all 10 triples; (ii) the 435 $\mathbb{G}_a^{3N(m)}$ translation generators across the 10 pairs $(m,k) \le 5$ all satisfy the sector-locality survival rule $\Phi_{m,k}(e^{(m)}_g \cdot x) = e^{(k)}_g \cdot \Phi_{m,k}(x)$ when the generator's sector has $n \le k$, and the action drops to zero when $n > k$; (iii) the class-level U^*-action on χ, η (depth-0 trivial by v3.66.0 FO3 Interpretation C) commutes with $\Phi_{m,k}$ bit-exact at all 10 pairs (40 identities). The SL_2 factor of $U^* = \mathbb{G}_a \rtimes SL_2$ acts on the Peter–Weyl decoration ($j_{\max}$ axis), independent of the $n_{\max}$ axis on which $\Phi_{m,k}$ acts; commutativity is categorical (independence-of-axes), not requiring a per-cell check. The Hopf-axiom compatibility under $\Phi_{m,k}$ is structurally trivial on the abelian primitive substrate of v3.61.0 Track A — the substantive PS-2 content lives at the cofiltered axiom + generator compatibility + class-action levels, which is the honest reading of why PS-2 is lighter than PS-1. Total bit-exact zero residuals across the panel: $10 + 435 + 40 = 485$. The cosmic-Galois U^* lifts to the PS-1 pro-system by construction; PS-3 carries the inverse limit $\mathcal{O}_\infty = \varprojlim \mathcal{O}_{n_{\max}}$ and extends the U^*-action to it, including the non-trivial part of Interpretation C on the continuum $F(s)$ Mellin lift (M2 Tate subgroup action + M3 standard motivic Galois action on $\mathrm{MT}(\mathbb{Q},1)$ odd-zeta classes per v3.66.0 FO2).

Pro-system lockdown PS-3: inverse limit $\mathcal{O}_\infty$ and the continuum $F(s)$ panel (Sprint Q5'-ProSystem-Lockdown, PS-3 sub-track, 2026-06-06; memo `debug/sprint_q5p_ps3_inverse_limit_memo.md`; data `debug/data/sprint_q5p_ps3_inverse_limit.json`). The truncated CH pro-system has a sequential inverse limit $\mathcal{O}_\infty = \varprojlim_{n_{\max}} \mathcal{O}_{n_{\max}} \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$ where $\mathbb{N}_{\mathrm{sec}} = \{(n,l) : n \ge 1, 0 \le l \le n\}$ is the infinite sector index, equipped with the product / inverse-limit topology under which every projection $\pi_{n_{\max}} : \mathcal{O}_\infty \to \mathcal{O}_{n_{\max}}$ is continuous. The continuum cocycle classes χ_∞, η_∞ are defined by the v3.60.0 closed-form sector-local extensions $\chi_{(n,l)} = +2$ if $l < n$ and $-2n$ if $l = n$; $\eta_{(n,l)} = (2l+1)(2n+1)$ if $l < n$ and $n(2n+1)$ if $l = n$. Universal property bit-exact at 6 cutoffs: $\pi_{n_{\max}}(\chi_\infty) = \chi^{(n_{\max})}$ and $\pi_{n_{\max}}(\eta_\infty) = \eta^{(n_{\max})}$ for every $n_{\max} \in \{1, \ldots, 6\}$ (154 sector-level identities, including the new falsifier cell $n_{\max} = 6$ with 27 sectors at shell 6 and $\dim \mathcal{H} = 224$). Continuity under transitions bit-exact at 15 pairs $\times$ 2 characters (30 identities). The v3.66.0 FO2 $F(s)$ integer-s panel at $s \in \{6, 7, 8, 9, 10\}$ (25 terms) carries to the inverse limit with bit-exact $\mathrm{MT}(\mathbb{Q},1)$ weight / depth grading per term: each M2 component $\pi^{2k} \cdot \mathbb{Q}$ has depth 0 weight $2k$, each M3 component $\zeta(2k+1) \cdot \mathbb{Q}$ has depth 1 weight $2k+1$; classification, weight, depth and $\mathrm{MT}(\mathbb{Q},1)$ membership all bit-exact at 25/25 terms by independent symbolic parsing. The cosmic-Galois U^* acts on the M2 / M3 components at the period level via Tate subgroup (M2) and standard motivic Galois action on odd-zeta generators (M3, Brown 2012, Glanois 2015); U^*-orbit closure at the panel's (weight, depth) level is automatic by the one-dimensionality of the (depth-1, weight-$2k+1$) slots in $\mathrm{MT}(\mathbb{Q},1)$ at weight ≤ 10. Total bit-exact zero residuals across the PS-3 panel: $154 + 30 + 100 = 284$. Three of four Pro-System-Lockdown sub-tracks now closed; PS-4 (endomorphism rigidity / Tannakian readiness probe) is the lockdown sprint's final sub-track.

Pro-system lockdown PS-4: endomorphism rigidity probe and Tannakian-readiness gap-list (Sprint Q5'-ProSystem-Lockdown, PS-4 sub-track, 2026-06-06; memo `debug/sprint_q5p_ps4_endo_rigidity`; data `debug/data/sprint_q5p_ps4_endo_rigidity.json`). $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_n)$ — the $\mathbb{Q}$-linear endomorphisms of the truncated sector-idempotent algebra that admit a compatible family at every lower cutoff $k \le n_{\max}$ — is the block-lower-triangular subalgebra under the shell filtration $N(1) \subset \cdots \subset N(n_{\max})$, with closed-form dimension $\dim_{\mathbb{Q}} \mathrm{End}_{\mathrm{compat}}(\mathcal{O}_n) = \sum_{1 \le j \le i \le n_{\max}} (i+1)(j+1)$; bit-exact panel dim $= 19, 55, 125, 245$ at $n_{\max} = 2, 3, 4, 5$ via direct basis enumeration matching the closed form ($444 + 370 + 4 + 4 + 50 = 872$ bit-exact checks). This is strictly larger than the depth-0 scalar diagonal $\mathbb{Q}^{N(n_{\max})}$ and strictly smaller than the full $M_{N(n_{\max})}(\mathbb{Q})$; algebra-of-functions rigidity at finite cutoff is structurally weaker than scalar-diagonal rigidity because the algebra is commutative and the transitions are sector projections. The Tannakian-rigidity content (in the Deligne–Milne 1982 sense) lives one level up at $\mathrm{Rep}(\mathcal{H}_{\mathrm{GV}})$, where the abelian primitive coproduct and antipode $S(x) = -x$ make rigidity automatic on finite-dim modules. Tannakian-readiness gap-list across the four standard prerequisites (Deligne–Milne 1982): abelian category structure, symmetric monoidal structure, and rigidity are SATISFIED at finite cutoff with sprint-scale (bounded, combined) bookkeeping closures; the fiber functor's identification $\mathrm{Aut}^{\otimes}(\omega) \cong U^*_{\mathrm{GeoVac,Levi}} = \mathbb{G}_a^{3N(n_{\max})} \rtimes SL_2$ is the heart of Tannakian closure proper and is long-range, requiring (i) explicit construction of $\mathrm{Aut}^{\otimes}(\omega)$ via Tannakian reconstruction, (ii) matching with v3.63.0 L1's Levi decomposition, and (iii) verification against v3.66.0 FO3 Interpretation C at the period level on the inverse limit $\mathcal{O}_\infty$ (PS-3). PS-4 closes the fourth and final sub-track of the Pro-System-Lockdown sprint; the substrate for Tannakian closure as a standalone sprint or collaboration target is now fully named.

Tannakian closure foundation TC-1a: $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}})$ *is abelian (Sprint Q5'-Tannakian-Closure, TC-1a sub-track, 2026-06-06; memo* `debug/sprint_q5p_tc1a_abelian_memo.md`*; data* `debug/data/sprint_q5` The category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ of finite-dim rational representations of the v3.61.0 Track A abelian primitive Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}}) = \mathcal{O}(\mathbb{G}_a^{3N(n_{\max})})$ is an abelian category in the Deligne–Milne 1982 sense. An object is a finite-dim $\mathbb{Q}$-vector space M with $3N(n_{\max})$ pairwise commuting nilpotent endomorphisms $\{X_g\}$ indexed by the primitive generators $g = (n, l, s)$; a morphism is a $\mathbb{Q}$-linear map intertwining all X_g. Verified bit-exact on a five-rep five-morphism panel at $n_{\max} \in \{2, 3\}$ across all six universal-property axioms: zero object (5 reps $\times$ 2 directions $\times$ 2 cutoffs), direct sum universal property (3 pairs $\times$ 5 identities $\times$ 2 cutoffs), kernel universal property (4 morphisms $\times$ 2 conditions $\times$ 2 cutoffs), cokernel universal property (4 $\times$ 2 $\times$ 2), mono = ker(coker) (2 $\times$ 3 $\times$ 2), epi = coker(ker) (2 $\times$ 3 $\times$ 2). Total bit-exact zero residuals: 106/106. TC-1a closes the FIRST of four sprint-scale Tannakian-closure prerequisites named by PS-4 (v3.69.0): abelian category structure. TC-1b (symmetric monoidal), TC-1c (rigidity), TC-1d (fiber functor) are the remaining sprint-scale prerequisites; TC-1e (the first stone of the long-range wall: $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{GeoVac,Levi}}$ at $n_{\max} = 2$) opens the long-range arc. The category structure inherits from the v3.61.0 Track A abelian primitive substrate with no GeoVac-specific obstruction; the closure runs in 0.03 s wall time on a representative panel.

Tannakian closure foundation TC-1b: symmetric monoidal structure (Sprint Q5'-Tannakian-Closure, TC-1b sub-track, 2026-06-06; memo `debug/sprint_q5p_tc1b_monoidal_memo.md`*; data* `debug/data/sprint_q5p_tc1b_monoidal.json`*).* The category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is symmetric monoidal under $\otimes_{\mathbb{Q}}$ with the diagonal Hopf action $X_g^{M\otimes N} = X_g^M \otimes I_N + I_M \otimes X_g^N$ determined by the v3.61.0 Track A abelian primitive coproduct $\Delta(x_g) = x_g \otimes 1 + 1 \otimes x_g$. Unit object $\mathbf{1}$ = trivial 1-dim rep; unitors λ_M, ρ_M are the identity matrices on $\dim(M)$; associator $\alpha_{M,N,P}$ is the identity matrix in the canonical lex basis; braiding $\sigma_{M,N}$ is the swap-permutation matrix and is symmetric ($\sigma_{N,M} \circ \sigma_{M,N} = \mathrm{id}$). Verified bit-exact on a five-rep panel at $n_{\max} \in \{2, 3\}$ across six axiom panels totalling 56 identities: tensor diagonal action (4 pairs $\times$ 2 checks $\times$ 2 cutoffs = 16), tensor functoriality (3 tests $\times$ 2 cutoffs = 6), unitor intertwining (3 reps $\times$ 2 unitors $\times$ 2 cutoffs = 12), associator intertwining (2 triples $\times$ 2 cutoffs = 4), braiding intertwining + symmetric (3 pairs $\times$ 2 checks $\times$ 2 cutoffs = 12), and coherence diagrams pentagon + triangle + hexagon (3 $\times$ 2 cutoffs = 6). Total bit-exact zero residuals: 56/56. This closes the SECOND of four sprint-scale Tannakian-closure prerequisites named by PS-4 (v3.69.0). TC-1c (rigidity) and TC-1d (fiber functor) are next; TC-1e (long-range first stone) opens the heart of Tannakian closure.

Tannakian closure foundation TC-1c: rigidity (Sprint Q5'-Tannakian-Closure, TC-1c sub-track, 2026-06-06; memo `debug/sprint_q5p_tc1c_rigidity_memo.md`*; data* `debug/data/sprint_q5p_tc1c_r`
The category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is *rigid.* Every finite-dim rep V admits a dual $V^\vee = \mathrm{Hom}_{\mathbb{Q}}(V, \mathbb{Q})$ with the contragredient Hopf action $X_g^{V^\vee} = -(X_g^V)^T$ determined by the antipode $S(x_g) = -x_g$ of the abelian primitive Hopf algebra (v3.61.0 Track A). Evaluation $\mathrm{ev}_V : V^\vee \otimes V \to \mathbf{1}$ is the canonical pairing $\phi_i \otimes e_j \mapsto \delta_{ij}$ in the dual basis; coevaluation $\mathrm{coev}_V : \mathbf{1} \to V \otimes V^\vee$ sends $1 \mapsto \sum_i e_i \otimes \phi_i$. Both snake (zigzag) identities reduce to matrix identities in the canonical lex bases (where the associator and unitors are identity matrices, cf. TC-1b) and are verified bit-exact on a four-rep panel (T_1, T_2, J_2, J_3) at $n_{\max} \in \{2, 3\}$ across seven axiom panels totalling 50 identities: contragredient action (4 reps $\times$ 2 cutoffs = 8), evaluation intertwining (8), coevaluation intertwining (8), first snake (8), second snake (8), double dual $(V^\vee)^\vee = V$ as strict equality of endo data (8), and unit self-dual $\mathbf{1}^\vee = \mathbf{1}$ (2). Total bit-exact zero residuals: 50/50. *Structural surprise:* the double dual is *strict equality* of endomorphism data on the Track-A abelian substrate (because $S^2 = \mathrm{id}$ via $S(x_g) = -x_g$ and $(M^T)^T = M$); this should not be claimed beyond TC-1c, since a non-abelian extension of the Hopf substrate would introduce a non-trivial canonical isomorphism in place of strict equality. TC-1c closes the THIRD of four sprint-scale Tannakian-closure prerequisites named by PS-4 (v3.69.0).

Tannakian closure foundation TC-1d: fiber functor ω (Sprint Q5'-Tannakian-Closure, TC-1d sub-track, 2026-06-06; memo `debug/sprint_q5p_tc1d_fiber_functor_memo.md`*; data* `debug/data/sprint_q`
The forgetful functor $\omega : \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})) \to \mathrm{Vec}_{\mathbb{Q}}$ sending a representation $(M, \{X_g^M\}_g)$ to its underlying $\mathbb{Q}$-vector space M verifies the four Deligne–Milne 1982 fiber-functor properties bit-exactly at $n_{\max} \in \{2, 3\}$ on the representative panel $\{\mathbf{1}, T_2, J_2, J_3\}$ and morphism panel $\{f_1, f_2, f_3, f_4\}$ (mono / epi / identity / zero): twenty bit-exact zero-residual identities per cutoff (one unit + four $\otimes$-preservation + four kernel + four cokernel + three direct-sum + four faithfulness), totalling 40 identities at zero residual. Combined with TC-1a (abelian) and TC-1b (symmetric monoidal), ω is identified as a faithful exact $\mathbb{Q}$-linear symmetric $\otimes$-functor with $\omega(\mathbf{1}) = \mathbb{Q}$. At the implementation level ω is essentially the no-op forgetful functor — a rep IS its underlying $\mathbb{Q}$-vector space plus the Hopf-action data, so forgetting returns the rep's dim / matrix unchanged; naturality of $\otimes$-preservation is automatic because tensor products use Kronecker products in canonical lex basis. The Tannakian-closure precondition for $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ is met at the verified-precondition level; the motivic Galois group $\mathrm{Aut}^{\otimes}(\omega) = U^{*(n_{\max})}_{\mathrm{GeoVac}} = \mathbb{G}_a^{3N(n_{\max})}$ is the v3.61.0 Track A structural reading. TC-1d closes the FOURTH and last sprint-scale Tannakian-closure prerequisite named by PS-4 (v3.69.0); TC-1e (long-range first stone: explicit inclusion $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at $n_{\max} = 2$) opens the heart of Tannakian closure.

Tannakian closure foundation TC-1e: first stone of the long-range wall (Sprint Q5'-Tannakian-Closure, TC-1e sub-track, 2026-06-06; memo `debug/sprint_q5p_tc1e_aut_inclusion_memo.md`*; data* `debug/data/sprint_q5p_tc1e_aut_inclusion.json`*).* The pro-unipotent factor $\mathbb{G}_a^{3N(n_{\max})}(\mathbb{Q})$ of $U^*_{\mathrm{GeoVac,Levi}} = \mathbb{G}_a^{3N(n_{\max})} \rtimes SL_2$ maps explicitly into the natural $\otimes$-automorphisms of the fiber functor ω (TC-1d) via $\Phi : \mathbb{Q}^{3N(n_{\max})} \to \mathrm{Aut}^{\otimes}(\omega)$, $\Phi(t)(V) = \exp(\sum_g t_g X_g^V)$. Each $\eta_V(t) = \Phi(t)(V)$ is the truncated matrix exponential of a $\mathbb{Q}$-linear combination of the commuting nilpotent endomorphisms $\{X_g^V\}_g$. All four Deligne–Milne 1982 natural $\otimes$-automorphism axioms (invertibility, unit, naturality, $\otimes$-compatibility) plus the group law $\Phi(t_1 + t_2) = \Phi(t_1) \cdot \Phi(t_2)$ are verified bit-exact on a five-rep panel $(T_1, T_2, J_2, J_3, K_2)$ — including reps J_2, J_3 activating primitive generator $g_0 = (1, 0, 0)$ and K_2 activating $g_1 = (2, 0, 1)$ — at $n_{\max} \in \{2, 3\}$ across four test parameter values $(0, t^{(1)}_{g_0}, t^{(2)}_{g_1}, t_{\mathrm{mixed}})$. Total bit-exact zero residuals: 98/98 (49 per cutoff: invertibility $5 \times 4 = 20$ + unit $1 \times 4 = 4$ + naturality $2 \times 4 = 8$ + $\otimes$ $3 \times 4 = 12$ + group law 5). SL_2 commutativity at the inclusion level is categorical (independence-of-axes: SL_2 acts on the $j_{\max}$ axis, Φ on the $n_{\max}$ axis), not requiring a per-cell check. TC-1e closes the *inclusion direction* $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at finite cutoff; the *converse equality* $\mathrm{Aut}^{\otimes}(\omega) = U^*$ (full Tannakian closure proper, requiring the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ / PS-3) remains the long-range content. This is the FIRST STONE of the long-range wall named by PS-4 (v3.69.0): the substrate for Tannakian closure as a standalone sprint or collaboration target is now fully constructed at the verified-precondition level, and the explicit inclusion direction is bit-exact.

Tannakian closure foundation TC-1f: explicit $\mathbb{G}_a \times SL_2$ inclusion + full-panel injectivity (Sprint Q5'-Tannakian-Closure, TC-1f sub-track, 2026-06-06; memo `debug/sprint_q5p_tc1f_sl2_inclusion` data `debug/data/q5p_tc1f_sl2_inclusion.json`). The SL_2 factor of $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N(n_{\max})} \rtimes SL_2$ is realised on the Peter–Weyl panel $\{V_{\mathrm{triv}}, V_{\mathrm{fund}} = \mathbb{Q}^2, \mathrm{Sym}^2 V_{\mathrm{fund}} = \mathbb{Q}^3\}$ with explicit rep $\rho : SL_2(\mathbb{Q}) \to \mathrm{GL}(V_{\mathrm{PW}})$, and the same three structural identities TC-1e proved for $\mathbb{G}_a$ — invertibility, $\otimes$-compatibility, group law — are bit-exact for SL_2 across an $SL_2(\mathbb{Q})$ panel of five elements (identity, unipotent, torus, generic, Weyl involution). The Levi-decomposition compatibility $\mathbb{G}_a \times SL_2 = \mathbb{G}_a \cdot SL_2$ on the combined rep $V \otimes V_{\mathrm{PW}}$ is bit-exact by the Kronecker identity $(A \otimes I)(I \otimes B) = (I \otimes B)(A \otimes I)$ on 100/100 (parameter, g, V, V_{PW}) cells. Injectivity of $\Phi : U^*_{\mathrm{Levi}}(\mathbb{Q}) \to \mathrm{Aut}^{\otimes}(\omega)$ on the full primitive panel at $n_{\max} = 2$ is bit-exact: for each of the $3 \cdot N(2) = 15$ primitive generators g a 2-dim test rep V_g with $X_g^{V_g} = E_{12}$ and $X_h^{V_g} = 0$ for $h \neq g$ detects g uniquely, with $\eta_{V_g}(t) = I$ iff $t_g = 0$; predicate bit-exact on 255/255 (generator, parameter regime) cells across zero, single-generator, and generic three-generator test parameters. Total TC-1f residuals: 490/490 (block 1 135: invertibility 15, $\otimes$ 45, group law 75; block 2 100; block 3 255). This extends the v3.73.0 TC-1e inclusion from the $\mathbb{G}_a$ factor to the explicit $\mathbb{G}_a \times SL_2$ inclusion at finite cutoff with explicit witnesses on the full primitive panel.

Honest scope. The QSM / Bost–Connes / Greenfield–Marcolli–Teh lineage (Greenfield–Marcolli–Teh, arXiv:1305.5492 (2014); Connes–Marcolli–Ramachandran, arXiv:math/0501424 (2005); Connes–Marcolli book 2008 Ch. 4) is RULED OUT as a transport target on four independent structural grounds (type III vs. type II/I, σ-twisted vs. standard Connes axioms, arithmetic adelic input vs. combinatorial graph input, infinite vs. finite cutoff; see the Greenfield–Marcolli transport memo). Viable lineage: Connes–Marcolli motivic Galois (arXiv:math/0409306) + Marcolli–Tabuada noncommutative motives + Fathizadeh–Marcolli periods ([25]) + Deligne–Glanois cyclotomic descent ([23], [24]). Long-range mathematical research project on the viable lineage.

8 Conclusion

The transcendental content of GeoVac spectral data lies in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathbb{Z}[i, 1/2]$, indexed by the master Mellin engine slot $k \in \{0, 1, 2\}$. The classification places the GeoVac framework inside the published periods/motives program (Kontsevich–Zagier, Brown, Deligne, Glanois, Fathizadeh–Marcolli) in a precise way: each of GeoVac's empirically-observed transcendentals has a named motivic home inherited from the period-theoretic content of the corresponding sub-mechanism. The vertex-parity-as-χ_{-4} correspondence at the core of GeoVac's two-loop QED sector (Paper 28 Theorem 3) is the literal Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural Dirichlet series.

A The Bost–Connes / scaling-site substrate as arithmetic-side analog

The cosmic-Galois U^* thread of §7.6 sits on the *geometric* side of the Connes–Marcolli motivic Galois machinery: its substrate is the Fock-projected S^3 graph, its periods are cyclotomic mixed-Tate at level ≤ 4, and its transcendentals are classified by the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ at $k \in \{0, 1, 2\}$. A distinct but adjacent strand of the broader NCG program treats the *arithmetic* side of ζ directly: the Bost–Connes system [1], the Connes–Consani scaling site [2], and the recent Connes–Consani–Moscovici zeta spectral triples [4]. This appendix records, for completeness, the structural relationship between that strand and the GeoVac substrate, and notes its honest scope.

BC system as arithmetic substrate. Bost and Connes [1] constructed a quantum statistical mechanical (QSM) system whose partition function is $\zeta(\beta)$ and whose symmetry group recovers $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$. Its substrate is adelic (the finite adèles $\mathbb{A}_f$, the integral lattices $\widehat{\mathbb{Z}}$, the Hecke algebra of commensurability classes) and its natural objects are KMS states at varying inverse temperature β. This substrate is structurally orthogonal to GeoVac's combinatorial Fock-projected S^3 graph: where GeoVac's U^* acts on Mellin-moment-labelled data over a finite vertex set, the BC system acts on adelic commensurability classes via the Hecke convolution algebra. The categorical orthogonality is recorded as one of the four structural grounds on which the QSM / Greenfield–Marcolli–Teh [5] lineage is ruled out as a U^* transport target for GeoVac (§7.6 honest-scope paragraph; type III vs. type II/I, σ-twisted vs. standard Connes axioms, arithmetic adelic input vs. combinatorial graph input, infinite vs. finite cutoff).

Scaling site as topos-level generalisation. Connes and Consani [2, 3] subsequently lifted the BC construction to a topos-theoretic substrate, the *scaling site*, whose points carry the multiplicative action of $\mathbb{R}_+^\times$ and whose structure sheaf encodes the explicit formulas of Riemann–Weil. Connes, Consani, and Moscovici [4] consolidate this into "zeta spectral triples" that package the explicit formulas at the level of a single spectral triple. This is the current state of the art on the arithmetic side and is the natural reference framework for the BC–RH program.

Honest scope and relationship to GeoVac. The arithmetic content of ζ accessible to the BC / scaling-site / zeta-spectral-triple substrate is *not native to GeoVac*. GeoVac's RH-adjacent object is the Dirac spectral zeta $D(s)$ on the Fock-projected S^3 graph (CLAUDE.md §1.7 WH6), whose zero statistics are empirically GUE-like but which is structurally separated from classical Riemann ζ by three independent walls (Sprint 4 RH-N/O, Paper 29 §6): zeros not on a single critical line, no spectral-triple-natural functional equation, wrong Weyl law class. Engaging the BC–RH program in earnest would require rebuilding GeoVac's substrate on adelic / scaling-site arithmetic input, not on the combinatorial S^3 graph this paper has classified. That is a parallel long-range research arc in its own right; it is not a sub-sprint of the present cosmic-Galois U^* program. We record it here so that the relationship is on the record: GeoVac and the BC–RH program target adjacent corners of the NCG landscape, but the substrates do not transport, and the present paper's classification of GeoVac periods is structurally orthogonal to whatever the zeta-spectral-triple substrate classifies.

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